



DRIVING GROWTH THROUGH DATA: PRODUCT PERFORMANCE AND MARKETING ANALYSIS FOR ADVENTUREWORKS

BUSINFO 702

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1. Executive Summary

The goal of this report is to provide actionable insights for Adventure Works (AW) to enhance its product marketing strategy and drive sustained business growth. Our group focused on Promoting the Best Products by analysing both online and in-store sales to determine which products and product categories are most profitable and popular. This study is centred on helping AW target the right products for promotion, thereby optimizing revenue generation and profitability.

Key Findings:

1. Product Popularity:

- Bikes are the most popular product category, contributing the largest volume of sales across both online and in-store channels.
- Specific items such as the Sport-100 Helmet and AWC Logo Cap are frequently purchased together with complementary products like jerseys and gloves. This indicates strong customer demand for cycling-related items.
- Clothing and Accessories are particularly dominant in online sales, reflecting an opportunity for targeted promotions in digital channels.

2. Profitability:

- The Mountain-200 bike series is the most profitable, showing strong sales performance in both online and in-store channels.
- Products like Tires and Tubes exhibit the highest profit margins (63%), highlighting them as prime candidates for promotional campaigns.
- A bundling strategy of high-margin products (e.g., helmets) with lower-margin items (e.g., jerseys) is recommended to improve overall profitability.

3. Sales and Marketing Recommendations:

- AW should prioritize online sales by expanding digital marketing campaigns, particularly focusing on high-demand categories such as bikes and accessories.
- Bundle promotions should be developed for products that are frequently purchased together, such as helmets with jerseys, to maximize customer spending and clear slow-moving inventory.

4. Geographical and Seasonal Trends:

- Sales peak during Q3, which aligns with cycling season, presenting an opportunity for seasonal marketing efforts.
- The US market dominates sales, indicating that region-specific promotions could further drive growth.

2. Introduction

The selected focus area for the Data Warehouse (DW) design is ‘Promotion of Products’. The objective is to provide actionable insights to Adventure Works (AW) on how they can promote their most suitable products to sustain and accelerate business growth.

This focus area allows us to address key questions such as:

1. Which products are most/least popular, both in-store and online, and how does their popularity affect revenue and costs?
2. Which products are most/least profitable, both in-store and online, and how does profitability impact the overall business?
3. Based on product popularity and profitability, which products should AW target in their sales and marketing promotions to maximize growth?

Additional Business Questions:

1. Which products perform best in specific markets (countries/regions)?
2. Which products perform best during certain times of the year?

By answering these questions, the data warehouse will enable AW’s marketing department to make informed decisions about which products should be prioritized for promotion. This is crucial as it will help AW to not only sustain its current growth but also expand into new markets and increase its profitability.

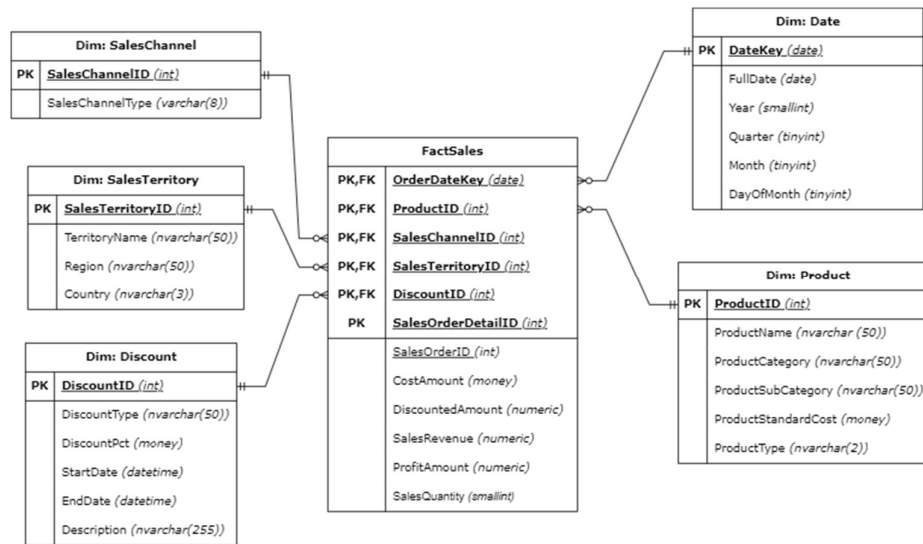
We chose Focus Area 2 because product performance is a critical driver of business success. By analysing product sales, profitability, and geographic performance, we can provide strategic recommendations that directly impact revenue growth. Additionally, this focus area aligns well with the available data in the AdventureWorks OLTP database, allowing us to build a robust star schema and ETL process to effectively support AW’s decision-making needs.

3. Data Warehouse Design

3.1 Star Schema Design

Figure 1

Star Schema Design



3.2 Description of Facts and Dimensions

The **FactSales** table stores transactional sales data, capturing key measures such as SalesQuantity, SalesRevenue, CostAmount, ProfitAmount, and DiscountedAmount. It also contains foreign keys that link to various dimension tables, enabling a comprehensive analysis of sales performance. Each row in the FactSales table represents a specific transaction, with connections to product, sales channel, sales territory, date, and discount details.

The **DimProduct** table is connected to the FactSales table via the ProductID foreign key. This allows sales data to be analysed by product attributes such as ProductName, ProductCategory, and ProductStandardCost, enabling insights into the performance of individual products.

The **DimSalesChannel** dimension is linked to the FactSales table through the SalesChannelID foreign key. This connection allows analysis of sales based on the channel type (SalesChannelType), providing insights into the differences in sales performance between online and offline transactions.

The **DimSalesTerritory** table is connected to the FactSales table via the SalesTerritoryID foreign key. This relationship enables geographical analysis of sales, allowing the business to break down revenue and profitability by region, territory, or country.

The **DimDiscount** table is linked to the FactSales table through the DiscountID foreign key. This connection provides the ability to analyse the effect of discounts on sales transactions, enabling an evaluation of the impact of different discount types and levels on revenue and profitability.

The **DimDate** dimension is connected to the FactSales table through the OrderDateKey foreign key. This connection supports time-based analysis, allowing sales performance to be tracked across different time periods, including day, month, quarter, and year.

4. ETL Design and Implementation

4.1 SSIS Project Overview

The SSIS project is designed to facilitate the extraction, transformation, and loading (ETL) of data from the AdventureWorks OLTP database into the data warehouse, ensuring accurate and clean data for reporting and analysis. The project extracts data from various sources such as sales orders, product, and date dimensions, and integrates them into fact and dimension tables.

As part of the ETL process, multiple transformations are applied to the data to ensure it is ready for analysis. This includes sorting and merging data from different sources (e.g., combining sales order headers with details, products with subcategories, etc.), creating derived columns to calculate new metrics such as SalesRevenue, CostAmount, and ProfitAmount, and replacing NULL values with placeholders such as "Unknown." Additionally, attributes like SalesChannelType are created based on flags within the source data, and certain fields are renamed for clarity and consistency in the data warehouse.

All transformations, including derived column creation, NULL value handling, renaming, and aggregation, are performed within the SSIS project as part of the ETL process, ensuring that the data is properly formatted, cleaned, and enriched before being loaded into the data warehouse for further analysis.

This approach ensures data integrity, supports efficient reporting, and enhances the overall analytical capabilities of the data warehouse.

4.2 ETL Process for FactSales

The FactSales table is the core fact table in the data warehouse, designed to capture detailed sales transactions. It integrates key data from orders, products, and time dimensions to support comprehensive sales analysis, including revenue, costs, and profitability.

4.2.1 Data Extraction:

- **SalesOrderHeader Source:**
Extracts order header information, including SalesOrderID, TerritoryID, OnlineOrderFlag, and OrderDate, totalling 31,465 rows. The selected columns capture essential order insights, including identifiers, geographic data, sales channels, and dates. These elements support sales performance analysis, strategic decisions, and operational improvements. They also provide key information like sales territory and order date,

facilitating the merge with order details and correlation with date dimensions for comprehensive sales analysis.

- **SalesOrderDetail Source:**
Extracts order detail data with fields such as SalesOrderID, SalesOrderDetailID, ProductID, OrderQty, UnitPrice, and LineTotal, totalling 121,317 rows. This details product information, quantities, and amounts for each order line, supporting sales revenue calculations.
- **Product Source:**
Extracts product cost data, including ProductID and StandardCost, totalling 504 rows. This information is used to append cost data to sales lines for profit calculations.
- **Date Source (Imported Flatfile):**
Extracts time dimension data, totalling 3,652 rows, providing time correlations to support daily, monthly, and yearly analysis.

4.2.2 Data Transformation:

- **Sort:** Each data source is sorted by specified columns to ensure correct data matching during merging or to facilitate data observation.
- **Merge Join:**
 - Merge SalesOrderHeader with SalesOrderDetail by SalesOrderID.
 - Merge Join 1: Further combines the merged order data with product cost data using ProductID, appending standard cost information to each transaction.
 - Merge Join 2: Connects the merged order data with date data via OrderDate to add time dimension information.
- **Derived Column:** New calculated columns - SalesRevenue, CostAmount, and ProfitAmount - are added to support sales and profit analysis. The expressions used to compute these columns are shown in Figure 2, which illustrates how key financial metrics are calculated to provide insights into profitability.

Figure 2

Derived Column Transformation for Profit, Discount, and Cost Amount Calculations

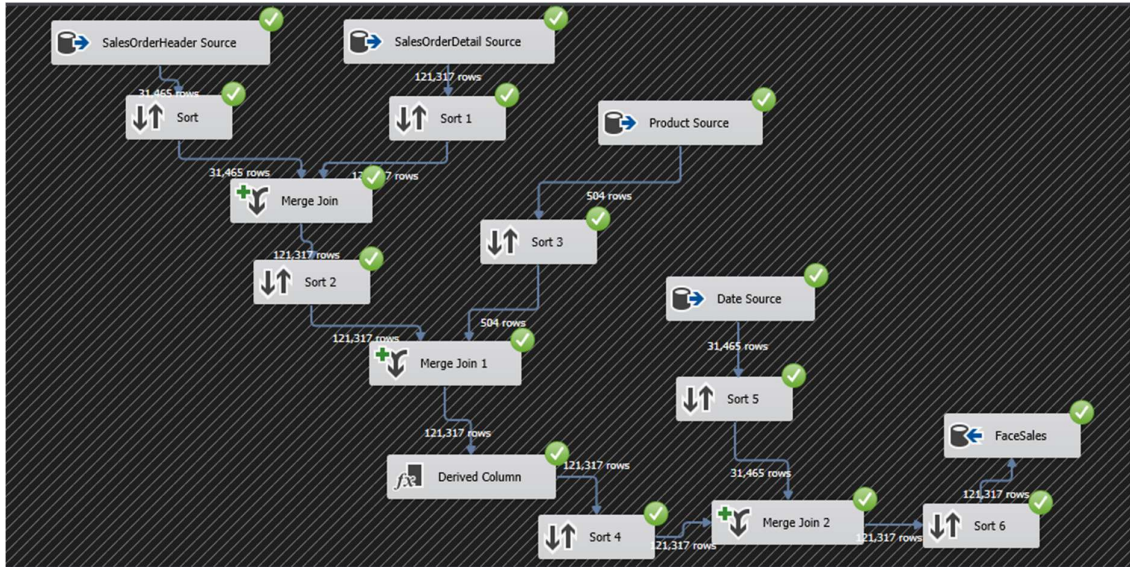
Derived Column Name	Derived Column	Expression	Data Type
ProfitAmount	<add as new column>	SalesRevenue - StandardCost * SalesQuantity	numeric [DT_NUMERIC]
DiscountAmount	<add as new column>	UnitPrice * SalesQuantity * DiscountPct	numeric [DT_NUMERIC]
CostAmount	<add as new column>	StandardCost * SalesQuantity	numeric [DT_NUMERIC]

4.2.3 Data Loading

The transformed and calculated data is loaded into the FactSales table (121,317 rows) in the data warehouse. Validation checks are performed by comparing the row counts to the raw data to ensure that no entries are missing or duplicated, and to verify the accuracy of key calculations (Refer to Figure 3).

Figure 3

ETL Flow for Loading Data into the FactSales Table



4.3 ETL Process for DimProduct Table

The DimProduct table is an essential dimension table in the data warehouse, capturing detailed product information that supports product performance analysis and profitability evaluations. This table integrates product, subcategory, and category data to provide comprehensive insights into product classification and costing.

4.3.1 Data Extraction:

- **Product Source:**
Extracts primary product data, including ProductID, ProductName, ProductSubcategoryID, StandardCost, and ProductType, totalling 504 rows. These columns provide key product details such as identifiers, names, costs, and categorization, which are crucial for understanding product-level performance and profitability.
- **ProductSubcategory Source:**

Extracts subcategory information, including ProductSubcategoryID, ProductCategoryID, and SubcategoryName, totalling 37 rows. This data helps classify products into subcategories, enhancing analysis by grouping similar items for better marketing management.

- **ProductCategory Source:**

Extracts category data, including ProductCategoryID and CategoryName, totalling 4 rows. This source links subcategories to broader product categories, supporting high-level categorization and analysis for strategic planning and decision-making.

4.3.2 Data Transformation:

- **Sort:** Each data source is sorted by relevant columns, such as ProductID and ProductSubcategoryID, to ensure correct data alignment during the merging process and to facilitate orderly data management. Sorting helps maintain data consistency and prepares the data for accurate merging.
- **Merge Join:**
 - Merge Product data with ProductSubcategory using ProductSubcategoryID, appending subcategory names to each product. This enhances classification and supports detailed product grouping for analysis.
 - Merge Join 1: Further merges the product-subcategory data with category information using ProductCategoryID, adding category details to each product. This step creates a fully integrated product view, enhancing the analytical capabilities of the data.
- **Derived Column:** The transformation involves replacing NULL values in existing columns rather than creating new ones. Specifically, ProductCategory and ProductSubcategory columns are updated to replace missing values with the term "Unknown," ensuring data completeness and consistency for analysis (See Figure 4).

Figure 4

Derived Column Transformation for ProductCategory, ProductSubcategory, and ProductType

Derived Column Name	Derived Column	Expression	Data Type	Length
ProductCategory	Replace 'ProductCateg...	(DT_WSTR,50)(ISNULL(ProductCategory) ? "Unkno...	Unicode string [DT_WS...	50
ProductSubcategory	Replace 'ProductSubca...	(DT_WSTR,50)(ISNULL(ProductSubcategory) ? "Unk...	Unicode string [DT_WS...	50
ProductType	Replace 'ProductType'	(DT_WSTR,50)ProductType	Unicode string [DT_WS...	2

- For ProductType, a different approach is taken due to replacement issues: the data type is first converted, and then NULL values are replaced with "Un" as a placeholder, ensuring that all records have consistent and usable values (See Figure 5).

Figure 5

Replacement Logic for ProductType with Data Type Conversion

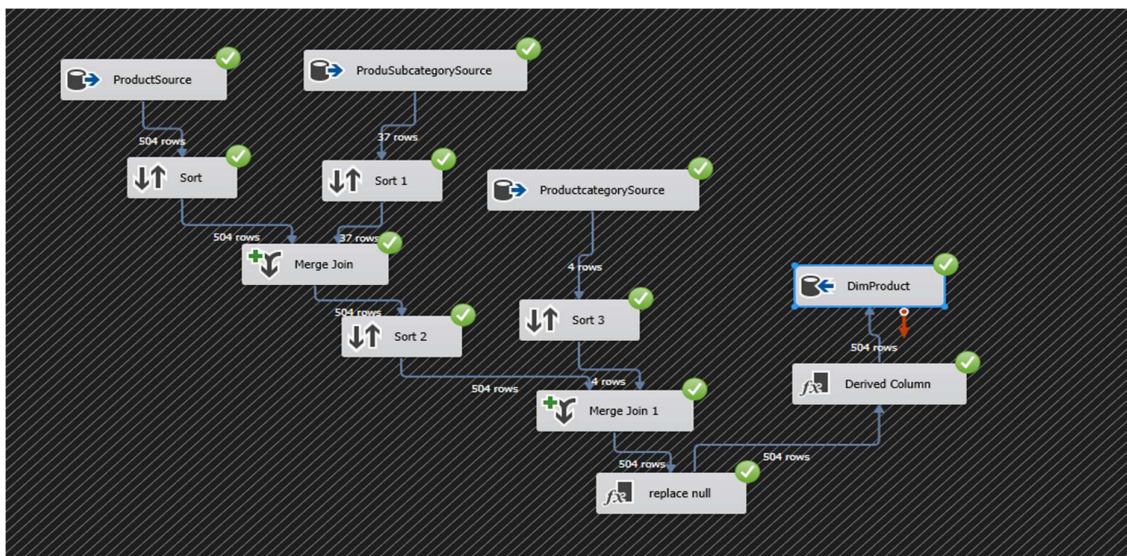
Derived Column Name	Derived Column	Expression	Data Type	Length
ProductType	Replace 'ProductType'	(DT_WSTR,50)(ISNULL(ProductType) ? "Un" : LEFT(P...	Unicode string [DT_WS...	2

4.3.3 Data Loading

The transformed data, consisting of 504 rows, is loaded into the DimProduct table in the data warehouse (See Figure 6). Validation checks compare row counts with the original source data to ensure accuracy, confirm no missing or duplicate entries, and verify the correctness of key calculations.

Figure 6

ETL Flow for Loading Data into the DimProduct Table



4.4 ETL Process for DimSalesChannel Table

The DimSalesChannel table captures information about different sales channels, specifically distinguishing between online and offline channels. This dimension is crucial for analysing sales performance across various channels and supports strategic decision-making in channel management.

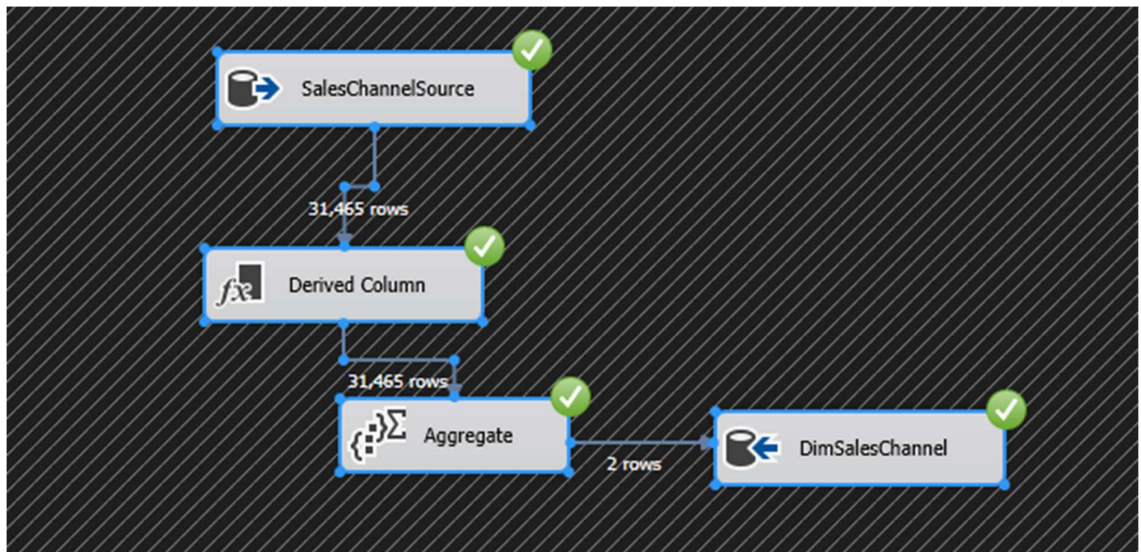
4.4.1 Data Extraction:

- SalesChannel Source:

Extracts sales channel data from the SalesOrderHeader table, specifically using the OnlineOrderFlag column, which indicates whether an order was placed online. This extraction includes a total of 31,465 rows, capturing the sales channel information for all orders (See Figure 7).

Figure 7

ETL Flow for Loading Data into the DimSalesChannel Table



4.4.2 Data Transformation:

- Derived Column:

- SalesChannelID: The SalesChannelID column directly applies the OnlineOrderFlag without modification.
- SalesChannelType: Another derived column is created to label the sales channel as either "Online" or "In-Store" based on the OnlineOrderFlag (See Figure 8). This transformation categorizes each order, enhancing the table's value in reporting and analysis.

Figure 8

Derived Column Transformation for SalesChannelID and SalesChannelType

Derived Column Name	Derived Column	Expression	Data Type
SalesChannelID	<add as new column>	OnlineOrderFlag	Boolean [DT_BOOL]
SalesChannelType	<add as new column>	(DT_WSTR,10)(OnlineOrderFlag == TRUE ? "Online...	Unicode string [DT_WS...

- **Aggregate:** Groups the data by SalesChannelID and SalesChannelType to aggregate channel information into distinct rows. The aggregation reduces the data from 31,465 rows to 2 rows, summarizing the sales channel types effectively (See Figure 9). This step helps consolidate sales channel data, making it more manageable and ready for analysis.

Figure 9

Aggregation of SalesChannelID and SalesChannelType

Input Column	Output Alias	Operation
SalesChannelID	SalesChannelID	Group by
SalesChannelType	SalesChannelType	Group by

4.4.3 Data Loading

The aggregated data, totalling 2 rows, is loaded into the DimSalesChannel table in the data warehouse. Validation checks are performed to ensure that the data is accurate and that the channels are correctly categorized without missing or duplicated entries. This validation ensures the integrity of the dimension and supports accurate channel-based sales analysis.

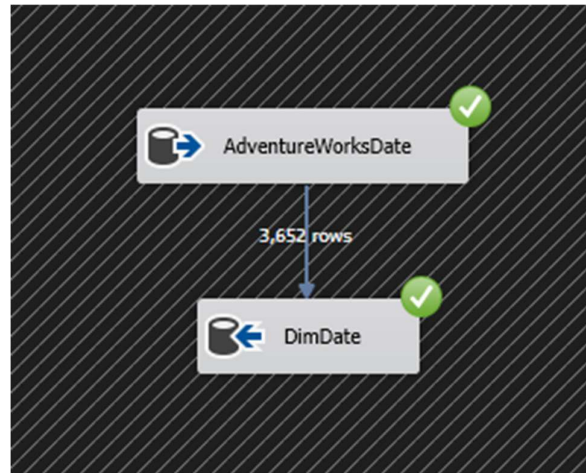
4.5 ETL Process for Remaining Dimension Tables

4.5.1 DimDate Table:

- **Data Source:** Extracts date information from the AdventureWorksDate source, containing 3,652 rows.
- **Columns Selected:** Includes key fields like DateKey, CalendarYear, CalendarQuarter, MonthNumberOfYear, DayNumberOfMonth, and FullDateAlternateKey (See Figure 10). These fields provide essential time-related information, supporting date-based analysis in the data warehouse.

Figure 10

ETL Flow for Loading Data into the DimDate Table

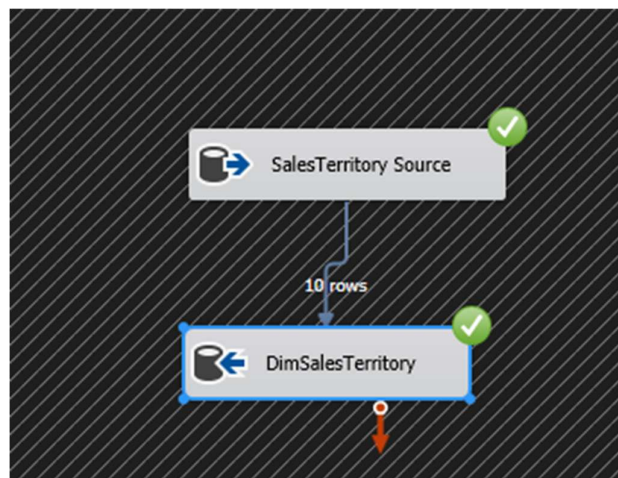


4.5.2 DimSalesTerritory Table:

- Data Source: Extracts sales territory information from the SalesTerritory Source, containing 10 rows.
- Columns Selected: Key fields include TerritoryID, Name, Group, and CountryRegionCode, which are renamed as SalesTerritoryID, TerritoryName, Region, and Country, respectively (See Figure 11). These fields enable geographic and regional analysis by linking sales data to specific territories.

Figure 11

ETL Flow for Loading Data into the DimSalesTerritory Table



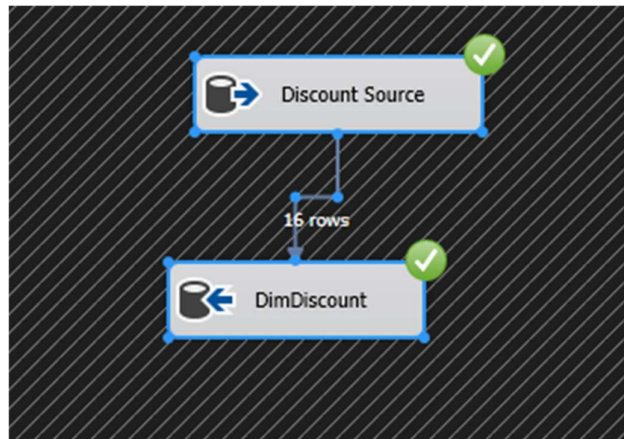
4.5.3 DimDiscount Table:

- Data Source: Extracts discount data from the Discount Source, containing 16 rows.

- Columns Selected: Key fields include SpecialOfferID, Type, DiscountPct, and Description. These columns provide detailed information about each discount, including its type, percentage, and description, which are critical for analyzing the impact of promotions on sales and profitability (See Figure 12).

Figure 12

ETL Flow for Loading Data into the DimDiscount Table



5. Cube Design

5.1 SSAS Project Overview

This report outlines the creation of a multidimensional cube for AdventureWorks' sales data analysis using SQL Server Analysis Services (SSAS). The primary goal of the SSAS cube is to provide a dynamic and detailed analysis tool for the company's sales, profitability, and discount performance. By utilizing the fact and dimension tables from the AdventureWorks Data Warehouse, the cube integrates key sales metrics and dimensions such as product categories, sales channels, and geographic regions.

Hierarchies within dimensions, including Date, Product, and Sales Territory, are implemented to support drill-down capabilities, allowing for in-depth analysis of sales trends over time, across products, and within specific regions. The cube's design facilitates high-level overviews as well as detailed explorations, making it a powerful tool for AdventureWorks' marketing department to drive data-driven decisions.

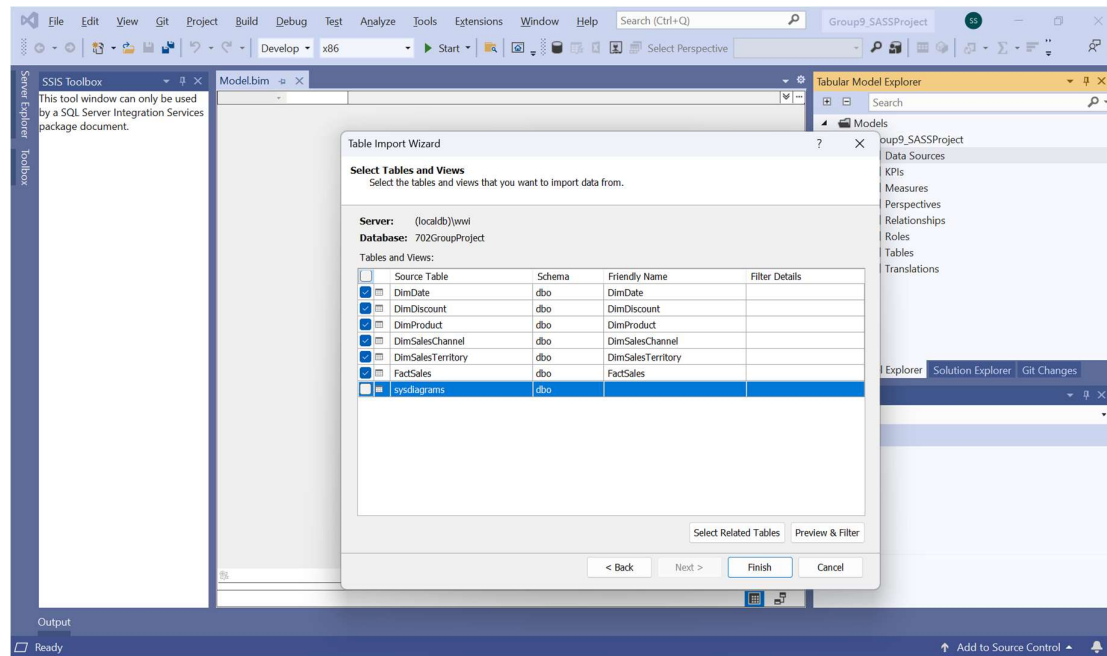
The measures in the cube include total sales revenue, profit margin, discount impact, and more, all of which provide critical insights into how different factors, such as promotions and sales channels, influence overall business performance. The well-structured cube allows for comprehensive business analysis, helping AdventureWorks identify growth opportunities and optimize sales strategies.

5.2 Data Source

The first step in the SSAS project was to import the Fact Table and Dimension Tables of the newly created Data Warehouse (DW) using SSIS. The tables were imported by connecting the "Data Sources" with the DW (See Figure 13).

Figure 13

Importing fact table and dimension tables from Data Warehouse as the source for the Cube



5.2.1 Fact Table: FactSales

- Description: Aggregates transactional data from the Sales.SalesOrderDetail and Sales.SalesOrderHeader tables. It includes detailed information on sales transactions such as quantities sold, revenue, cost, profit, and discounts.
- Key Fields: SalesOrderDetailID, SalesOrderID, ProductID, SalesTerritoryID, SalesChannelID, OrderDateKey, SalesQuantity, SalesRevenue, CostAmount, ProfitAmount, DiscountID, DiscountAmount.

5.2.2 Dimension Table

- DimDate: Contains date-related information useful for time-based analysis, including attributes like DateKey, FullDate, Year, Quarter, Month, and DayOfMonth.
- DimDiscount: Stores information about discounts, including DiscountID, DiscountType, DiscountAmount, StartDate, EndDate, and Description.
- DimProduct: Holds product details, such as ProductID, ProductName, ProductCategory, ProductSubcategory, ProductStandardCost, and ProductType.
- DimSalesChannel: Provides data on sales channels, including SalesChannelID and SalesChannelType (Online or In-Store).

- DimSalesTerritory: Includes sales territory information, such as SalesTerritoryID, TerritoryName, Region, and Country.

5.3 Cube Design

5.3.1 Overview of Cube Structure (Default Perspective)

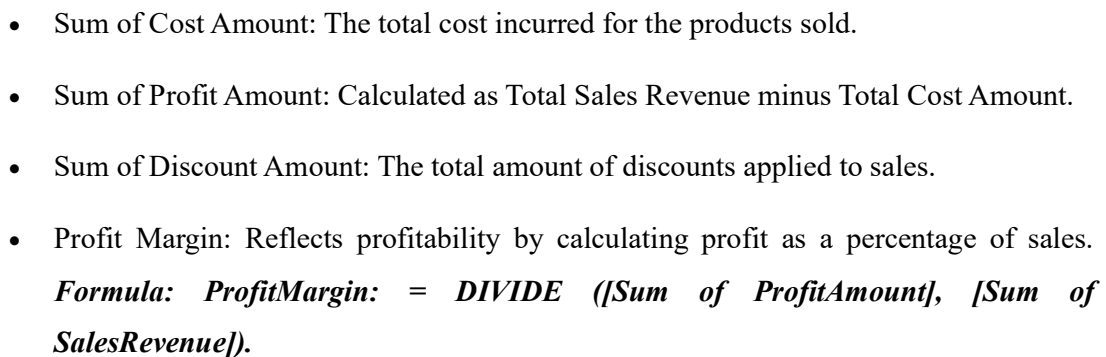
The cube designed for the Adventure Works Data Warehouse is a key component for analyzing sales performance, product popularity, and profitability. It integrates data from various fact and dimension tables to provide a comprehensive view of the company's operations. The cube taken into consideration has been kept as <Default> perspectives, with all the dimension tables from the DW, as it allows for a holistic approach to analyse the different metrics of the company's performance and identifying areas for potential growth in terms of its products. The cube is structured to support complex queries and facilitate insightful analysis for the marketing department.

5.3.2 Measures Creation in Cube

The cube includes several key measures derived from the FactSales fact table:

- Sum of Sales Quantity: Represents the total quantity of products sold.
- Sum of Sales Revenue: The total revenue generated from sales transactions, accounting for applicable discounts (See Figure 14).

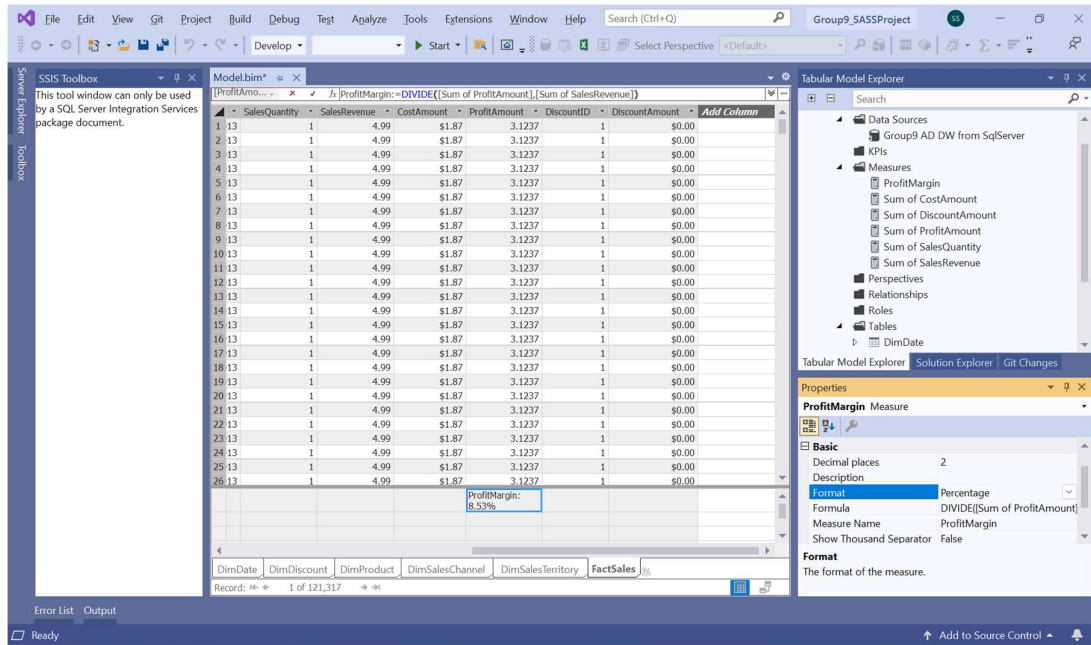
Calculation of Sum of Sales Revenue Measure



19

Figure 15

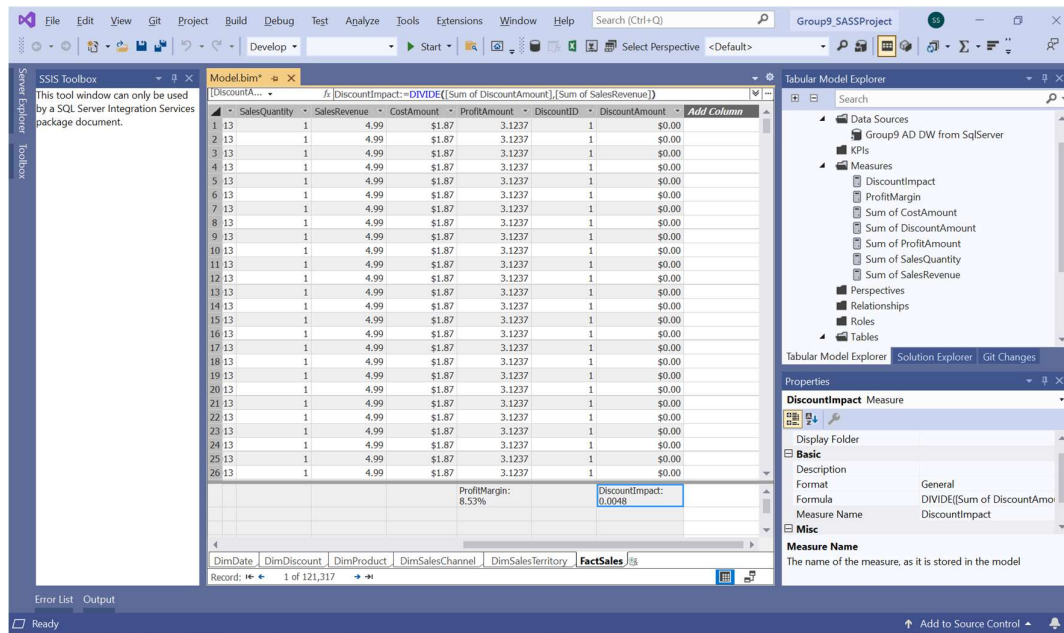
Calculation of Profit Margin Measure in Percentage



- **Discount Impact:** Measures the impact of discounts on overall sales, allowing the business to see how much revenue is affected by different discount rates.
Formula: DiscountImpact: = DIVIDE ([Sum of DiscountAmount], [Sum of SalesRevenue]) (See Figure 16).

Figure16

Calculation of Discount Impact Measure



The measures selected reflect core business metrics required for performance analysis. For instance, SalesRevenue directly ties to financial performance, while ProfitMargin helps assess how efficiently the company is operating. DiscountImpact provides insights into how pricing strategies are affecting the bottom line, which is crucial for any promotional analysis. Each of these measures is logically connected to business questions, such as:

- "Which products generate the most profit?"
- "How are discounts affecting sales?"
- "Which territories and channels are driving the most revenue?"

By focusing on these critical metrics, the cube is well-equipped to answer key business questions related to sales, profitability, and marketing strategies.

5.4 Relationship Creation in SSAS

In SQL Server Analysis Services (SSAS), establishing relationships between tables is a fundamental aspect of designing a functional cube. These relationships are configured within the Data Source View (DSV) and are critical for enabling accurate and meaningful data analysis. For the AdventureWorks cube, relationships are defined between the FactSales fact table and

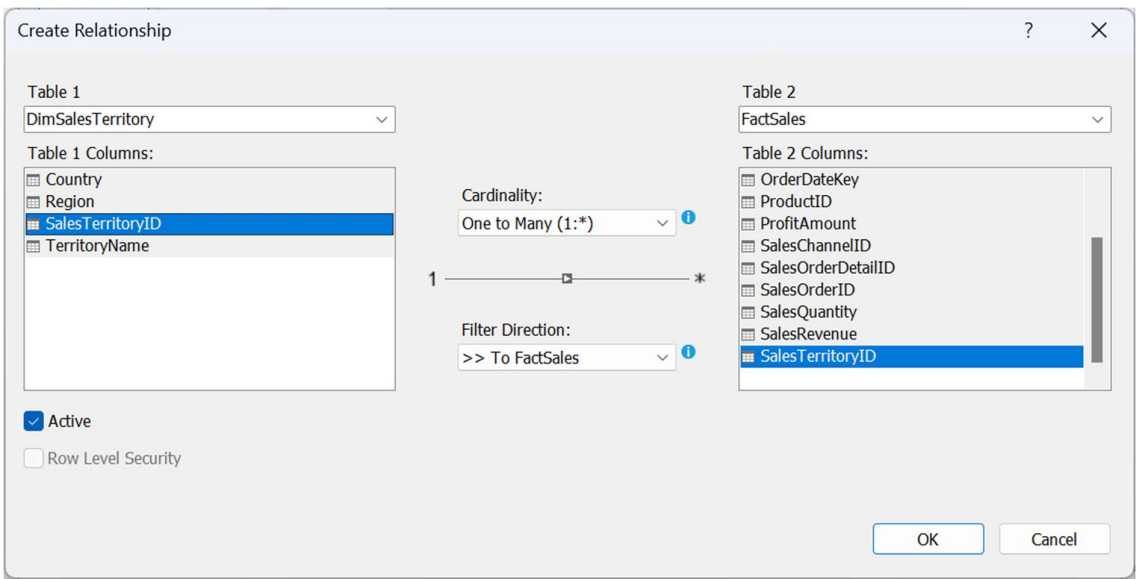
several dimension tables: DimDate, DimDiscount, DimProduct, DimSalesChannel, and DimSalesTerritory. These relationships leverage primary and foreign keys to ensure proper data linkage and integrity.

The Data Source View (DSV) integrates the FactSales table with these dimension tables through surrogate keys.

- SalesTerritoryID in DimSalesTerritory links to FactSales, enabling detailed analysis of sales performance by territory (See Figure 17).

Figure 17

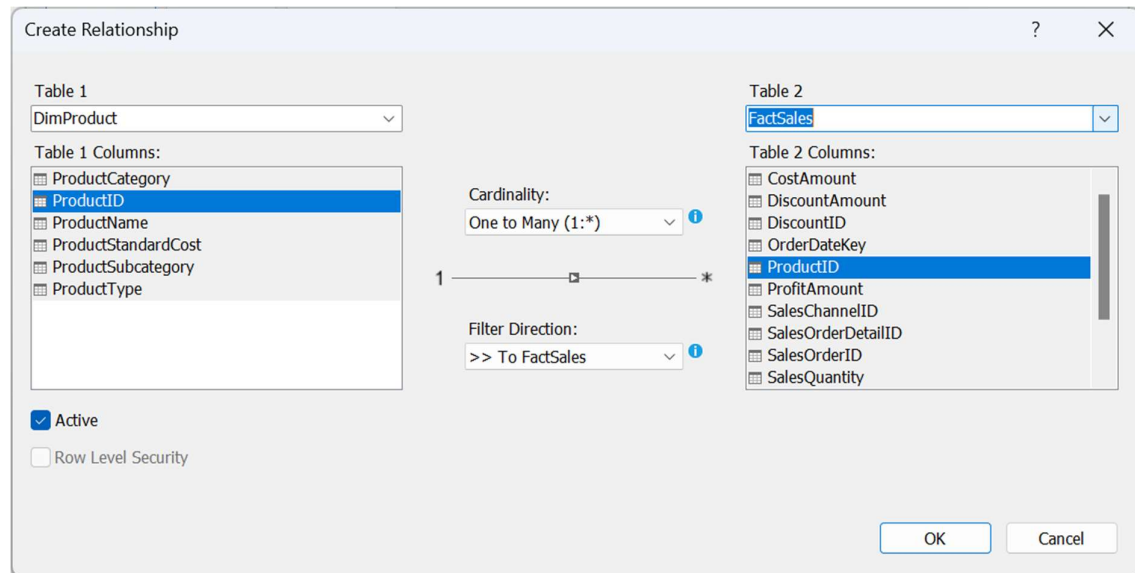
Relationship creation between DimSalesTerritory and FactSales



- ProductID in DimProduct connects to FactSales, facilitating analysis of sales by individual products or categories (See Figure 18).

Figure 18

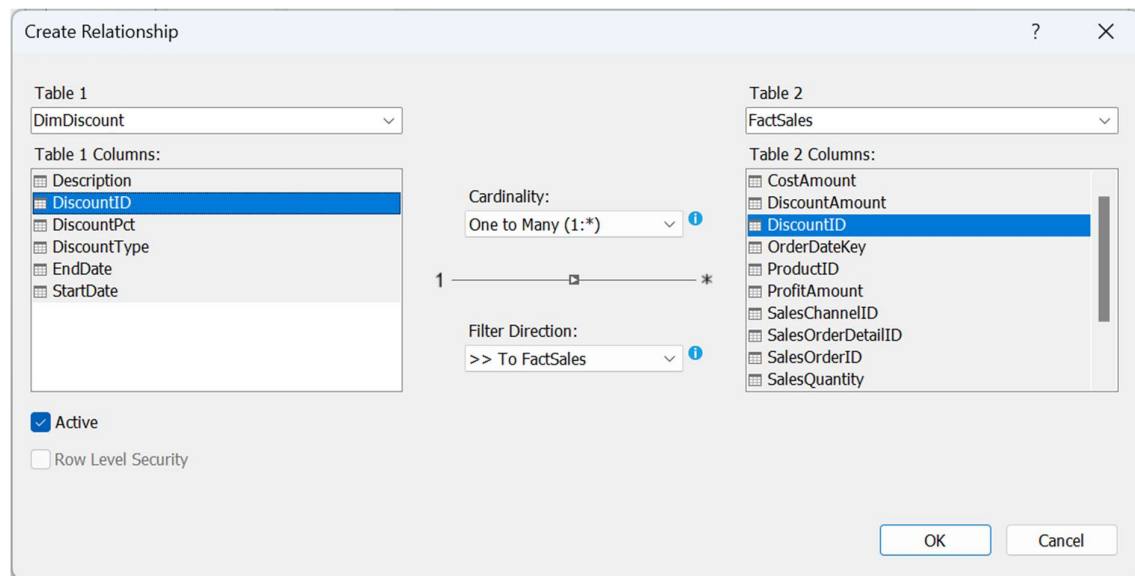
Relationship creation between DimProduct and FactSales



- DiscountID in DimDiscount ensures that the cube can track and analyse the impact of various discounts on total sales and profit (Figure 19).

Figure 19

Relationship creation between DimDiscount and FactSales



- DateKey in DimDate allows for temporal analysis of sales trends, enabling views by day, month, or year (See Figure 20).

Figure 20

Relationship creation between DimDate and FactSales

Create Relationship

Table 1
DimDate

Table 1 Columns:

- DateKey
- DayOfMonth
- FullDate
- Month
- Quarter
- Year

Cardinality:
One to Many (1:*)

Filter Direction:
>> To FactSales

Table 2
FactSales

Table 2 Columns:

- CostAmount
- DiscountAmount
- DiscountID
- OrderDateKey
- ProductID
- ProfitAmount
- SalesChannelID
- SalesOrderDetailID
- SalesOrderID
- SalesQuantity

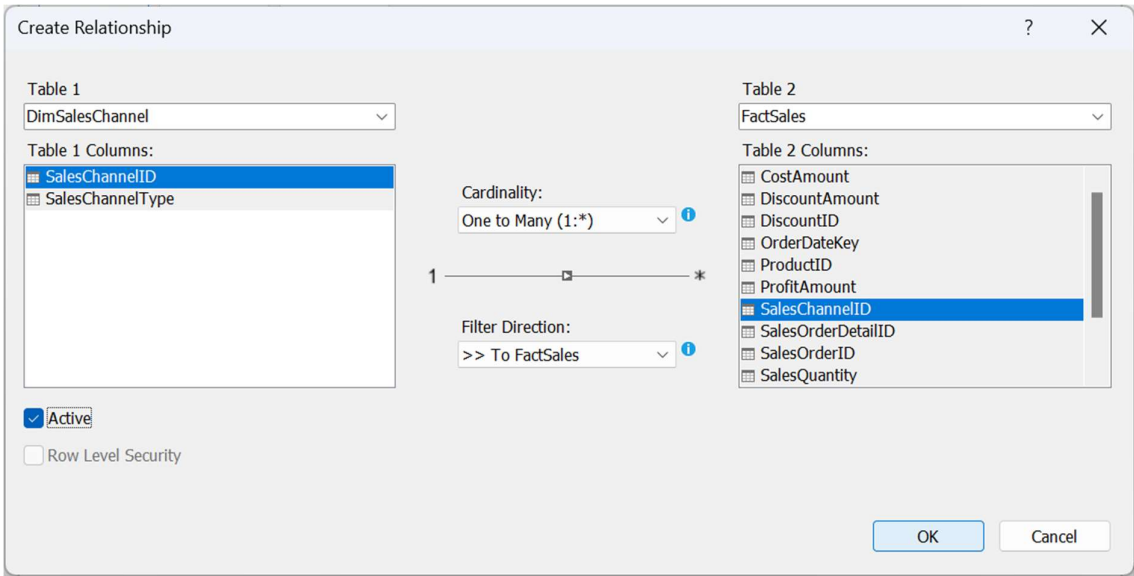
☒ Active

☐ Row Level Security

OK Cancel

- SalesChannelID in DimSalesChannel associate's sales activities with specific channels, providing insights into which channels (Online or In-Store) drive the most revenue or profit (See Figure 21).

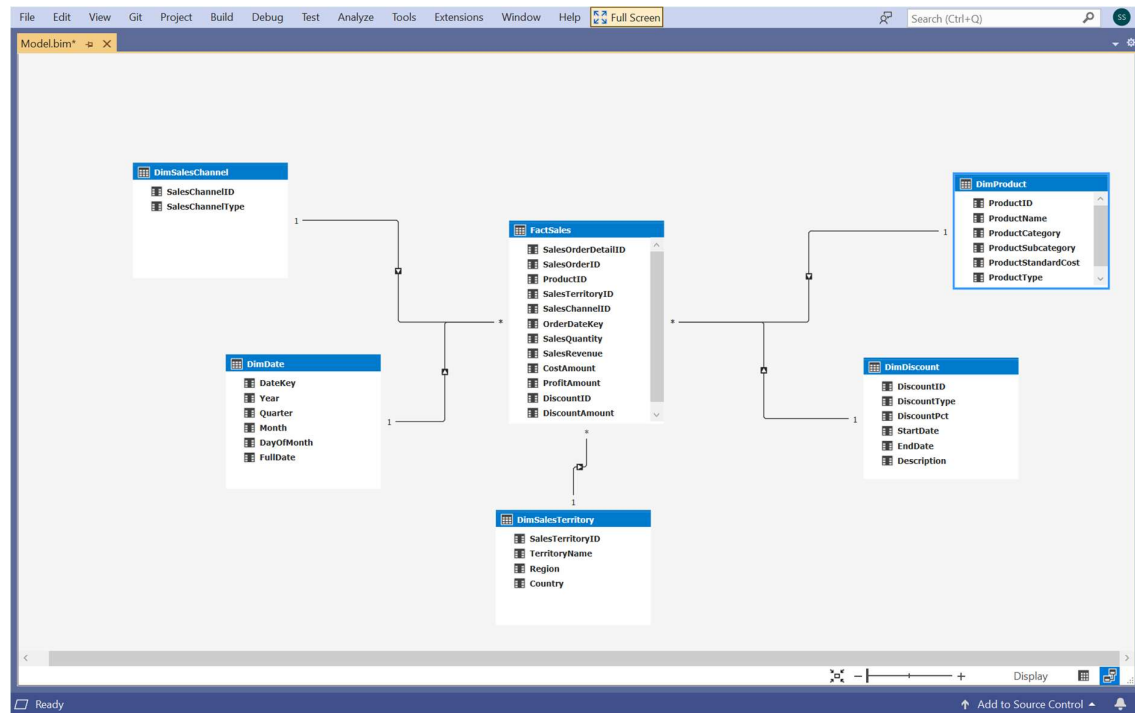
Figure 21
Relationship creation between DimSalesChannel and FactSales



These relationships ensure that each entry in the FactSales table can be analysed across multiple business dimensions such as product categories, sales channels, and geographical regions (See Figure 22). This setup aligns with the star schema, where FactSales serves as the central fact table and the dimension tables provide descriptive attributes for in-depth analysis.

Figure 22

Final Relationship Matrix of the Cube



The DSV setup abides to the principles of OLAP cube design, where each dimension connects through primary-to-foreign key relationships. This robust relationship mapping supports efficient querying, accurate data aggregation, and scalable performance for business analysis. By creating well-defined relationships, the cube is equipped to handle large volumes of data and deliver actionable insights effectively.

5.5 Dimension Design and Hierarchies

Dimension design and hierarchies are essential parts of creating a useful SSAS cube. Dimensions give context to the facts in the fact table, while hierarchies allow us to analyze data at different levels of detail. This makes the data more flexible and easier to explore.

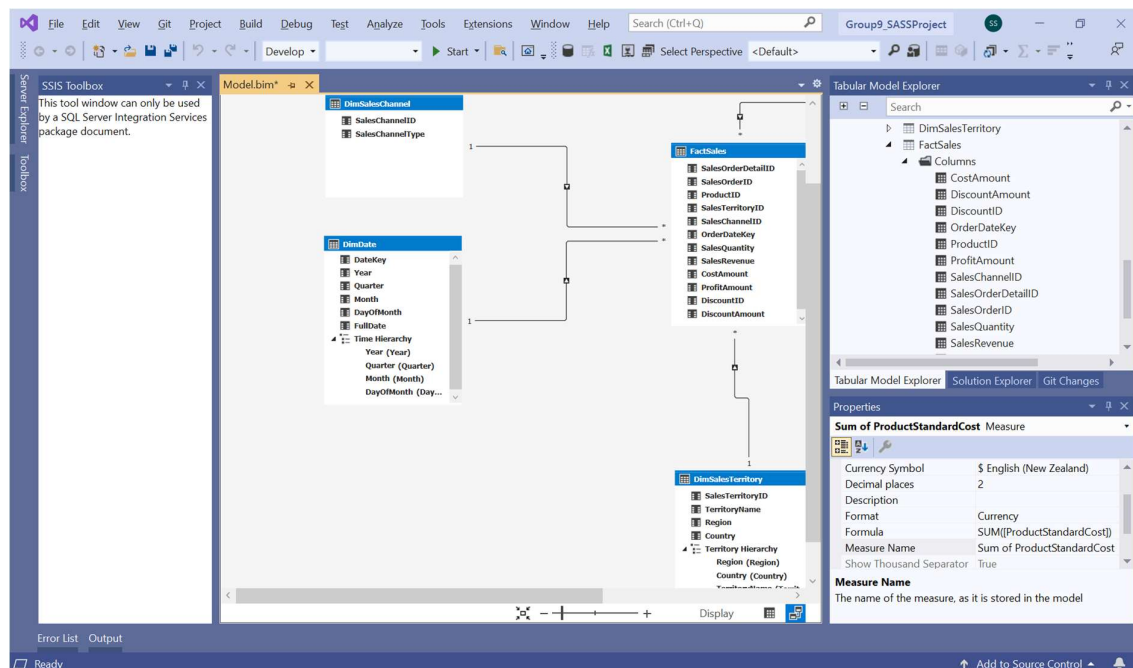
5.5.1 Hierarchies in Dimension

Hierarchies are like layers of data that help us drill down into more specific details or roll up to broader summaries. Here's how the hierarchies are set up in the Adventure Works cube:

- **Date Hierarchy:** In the DimDate dimension, the Time Hierarchy is organized as Year > Quarter > Month > DayOfMonth (See Figure 23). This lets users look at sales data from a yearly overview down to daily details. It's great for spotting seasonal trends and tracking changes over time.

Figure 23

Creation of Time Hierarchy in DimDate

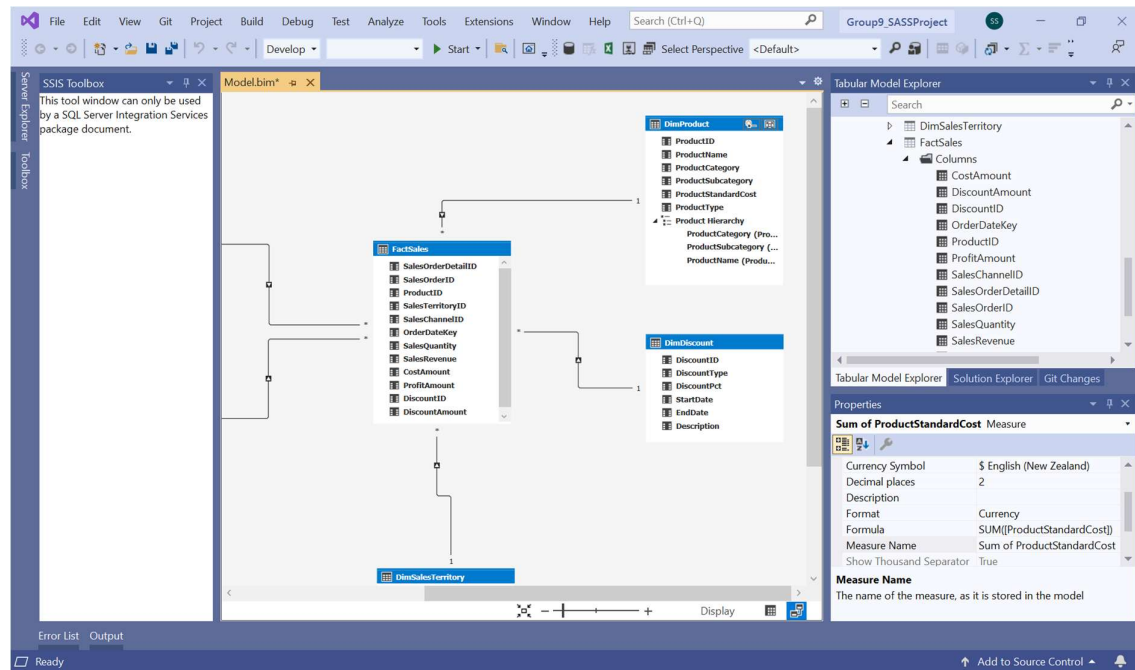


- **Product Hierarchy:** For the DimProduct dimension, the Product Hierarchy is ProductCategory > ProductSubcategory > ProductName (See Figure 24). This setup helps users to analyse data starting from broad product categories and drill down to

specific products. It's useful for comparing different products and understanding their performance within categories.

Figure 24

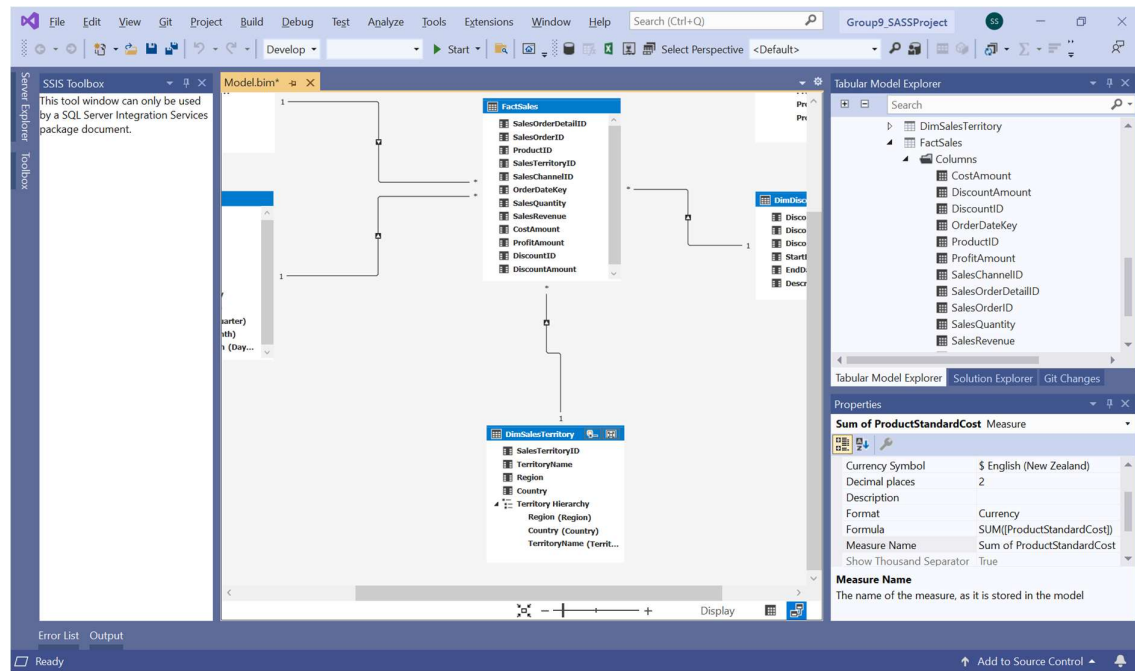
Creation of Product Hierarchy in DimProduct



- **Geographical Hierarchy:** In the DimSalesTerritory dimension, the Territory Hierarchy is Region > Country > TerritoryName (See Figure 25). This allows users to look at sales data across different geographic levels, from countries down to specific territories.

Figure 25

Creation of Territory Hierarchy in DimSalesTerritory

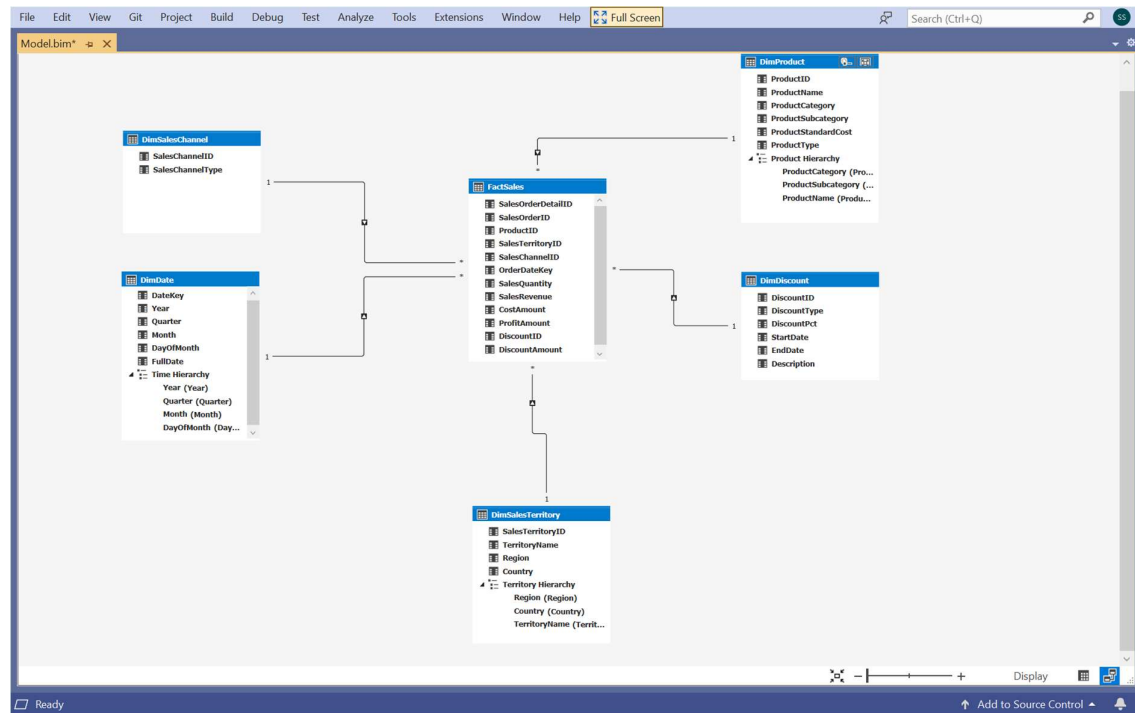


5.6 SSAS Cube Data Model and Relationships

In conclusion, the design and implementation of the SSAS cube for the Adventure Works project provide a robust framework for analysing and reporting data. By effectively creating and linking dimensions such as Date, Product, Discount, Sales Channel, and Sales Territory with the FactSales table (See Figure 26), the cube supports comprehensive data analysis. The hierarchical structures within these dimensions enhance the ability to drill down and roll up through various levels of detail, facilitating more nuanced insights.

Figure 26

Final Diagram View of the Cube



Integration with Microsoft Excel further enriches the cube's functionality, allowing users to utilize PivotTables, slicers, and hierarchical navigation to explore data in a user-friendly manner. This integration not only simplifies complex analyses but also supports detailed reporting and decision-making processes.

Overall, the SSAS cube's thoughtful design and seamless Excel integration ensure that users can perform thorough and flexible data analysis, driving better business insights and strategic decisions for Adventure Works.

6. Cube Exploration and Analysis

6.1 Introduction to Cube Browsing

This section demonstrates how the SSAS cube can be used to answer key business questions by utilizing the "Analyse in Excel" feature. The cube was exported from SSAS into excel and Pivot Table functionality was used to explore and develop an understanding for the first question.

This functionality allows stakeholders to interact directly with the cube data in Excel, enabling them to filter, explore, and drill down into detailed insights. By browsing the cube, stakeholders can access critical information related to product popularity, profitability, and marketing performance, helping them make informed, data-driven decisions.

The following subsections outline how the cube's dimensions and measures are utilized in Excel to address three key questions related to product performance. Screenshots of the different Excel analysis are provided to illustrate the process, followed by a concise summary of the results and a justification based on the data. This approach highlights the practical benefits of using SSAS cubes with Excel for dynamic and flexible data analysis.

6.2 Question1: Which products are most/ least popular either instore or online and how does popularity impact revenue/cost of the products?

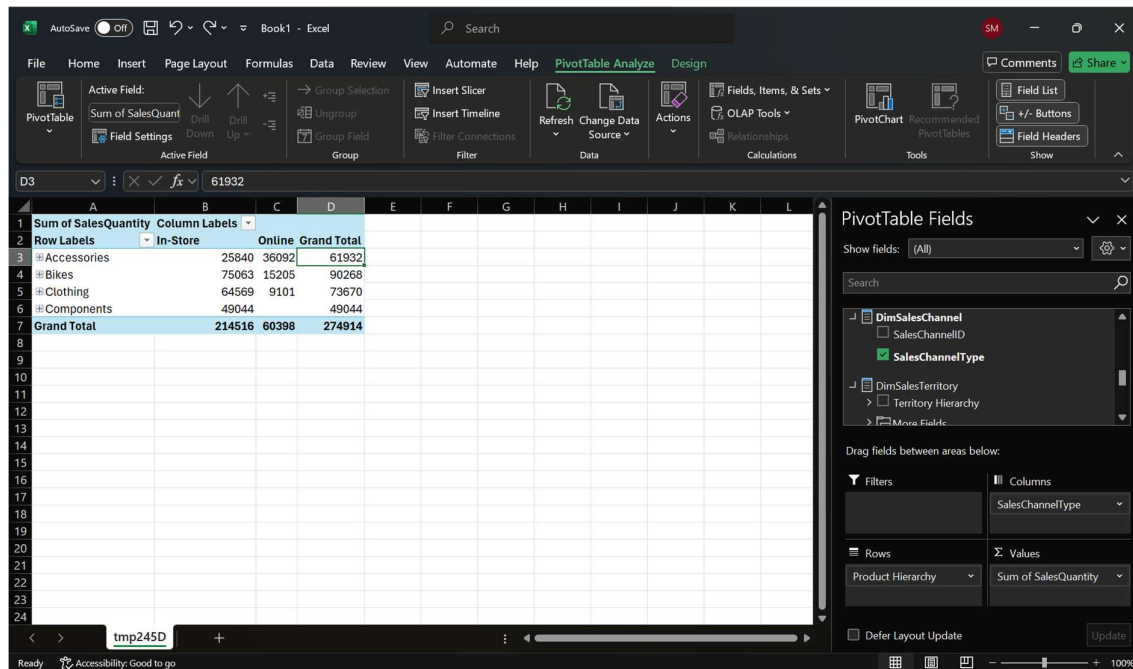
6.2.1 Cube Usage

The question was broken down to be analysed in 3 parts. For answering this question, "Popularity" has been defined by metric of Quantity of Goods Sold (Sum of SalesQuantity).

1. Product Popularity Analysis:
 - Setup: The analysis began by setting the columns to SalesChannelType (In-Store and Online) and rows to the Product Hierarchy, with the values showing the Sum of SalesQuantity (See Figure 27).
 - Drill-Down: Product Hierarchy was drilled down to Product Name.
 - Sorting: Products were sorted by Sales Quantity to identify the most popular (largest to smallest) and least popular (smallest to largest) items across different sales channels.

Figure 27

Pivot Table selection and results for Popularity Analysis (Product Categories)

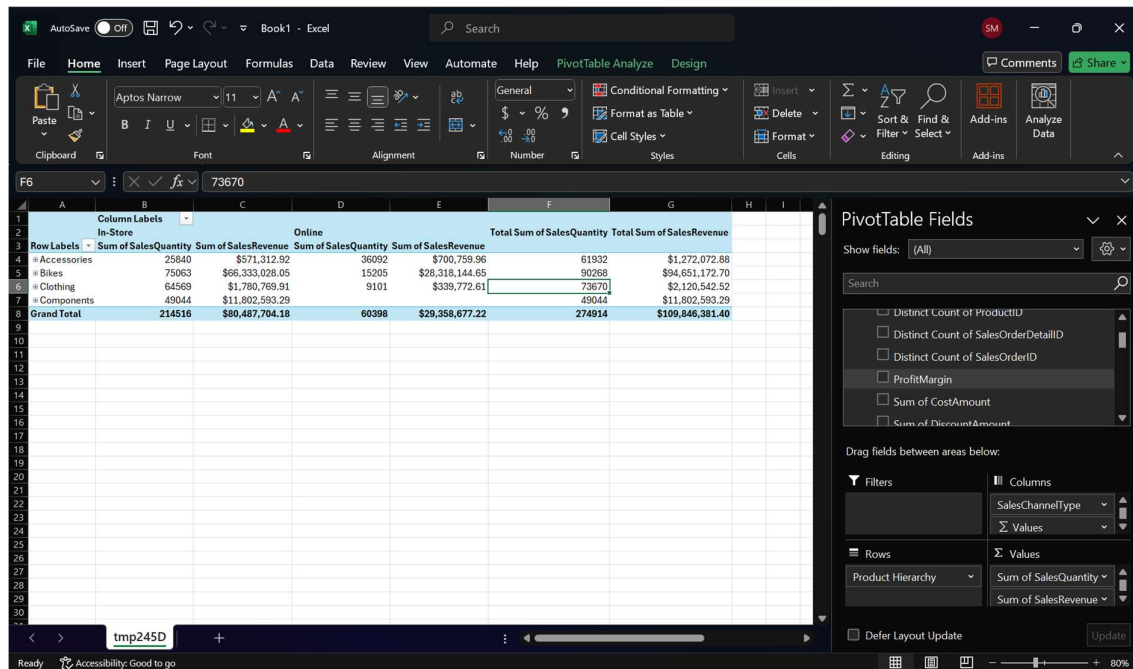


2. Impact on Revenue and Cost:

- Setup: A new PivotTable was configured with columns for SalesChannelType, rows for Product Name, and values for both Sum of SalesQuantity and Sum of SalesRevenue (See Figure 28). Sum of CostAmount is used to analyse the impact on revenue and cost.

Figure 28

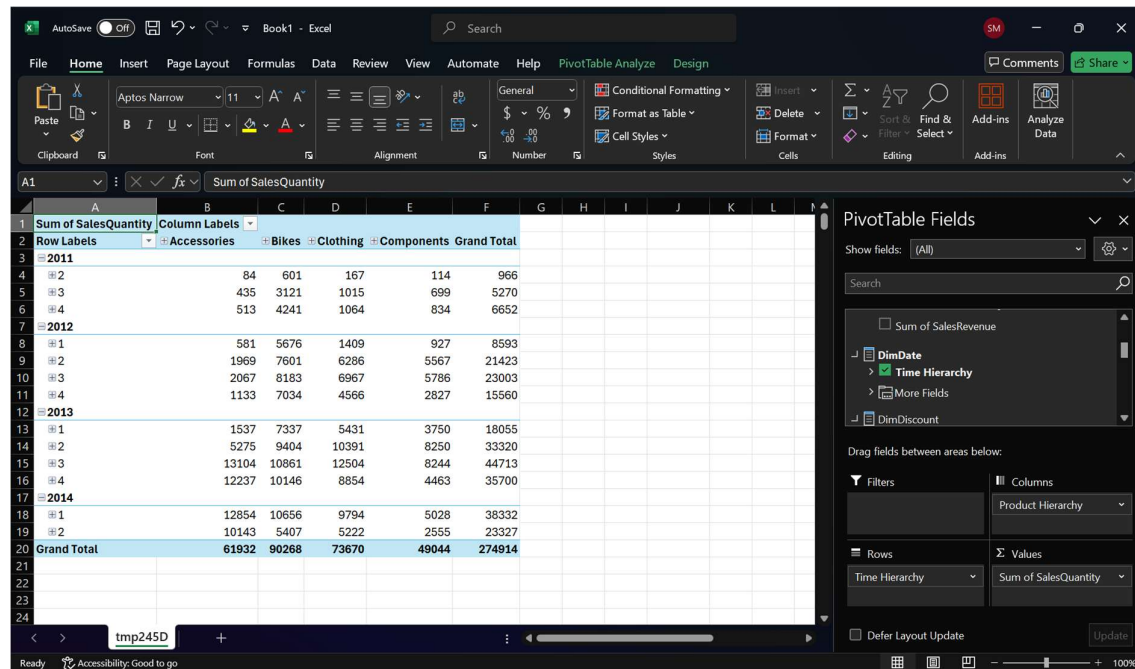
Pivot Table selection and results for Impact Analysis on Qty Sold (Sales Revenue)



- Charts: Combo charts compared Sales Quantity with Revenue and Cost to see how popularity affects these metrics.
3. Trends Over Time and Location:
- Time Analysis: For time trends, the columns were set to the Product Hierarchy and rows to the Time Hierarchy, with values for Sum of SalesQuantity. A Line with Markers chart showed trends by Quarter (See Figure 29).

Figure 29

Pivot Table selection and results for Time Analysis on Qty Sold (Product Categories)



- **Location Analysis:** For geographical insights, the columns included the Product Hierarchy, rows were set to the Territory Hierarchy (drilled down to Country), and values were Sum of SalesQuantity. A Clustered Column chart illustrated sales across different countries.

6.2.2 Findings

In terms of overall popularity, measured by sales quantity, "Bikes" are the most sought-after category (See Table 1).

Table 1

Popularity by category

Sum of SalesQuantity	Column Labels		
Row Labels	In-Store	Online	Grand Total
Bikes	75,063	15,205	90,268
Clothing	64,569	9,101	73,670
Accessories	25,840	36,092	61,932
Components	49,044		49,044
Grand Total	214,516	60,398	274,914

Drilling down to specific products, "Accessories" dominate online sales, with the "Water Bottle - 30 oz." leading in popularity (See Table 2).

Table 2*Online most popular product*

Sum of SalesQuantity	Column Labels	
Row Labels	Online	Grand Total
Water Bottle - 30 oz.	4244	4244
Patch Kit/8 Patches	3191	3191
Mountain Tire Tube	3095	3095
Road Tire Tube	2376	2376
Sport-100 Helmet, Red	2230	2230

In-store "Clothing" products like the "Long-Sleeve Logo Jersey, L" are highly flavored (See Table 3).

Table 3*In-store most popular product*

Sum of SalesQuantity	Column Labels	
Row Labels	In-Store	Grand Total
Long-Sleeve Logo Jersey, L	6,140	6,140
AWC Logo Cap	6,121	6,121
Sport-100 Helmet, Blue	4,618	4,618
Sport-100 Helmet, Black	4,447	4,447
Classic Vest, S	4,079	4,079

In contrast, "Components" products like the "HL Touring Frame - Blue, 50" are the least popular in-store (Table 4), reflecting niche demand for specialized items.

Table 4*In-store least popular product*

Sum of SalesQuantity	Column Labels	
Row Labels	In-Store	Grand Total
LL Touring Frame - Blue, 58	4	4
ML Mountain Frame-W - Silver, 38	7	7
LL Mountain Frame - Black, 40	8	8
LL Road Seat/Saddle	10	10
Classic Vest, L	12	12

Meanwhile, lower-end models like the "Mountain-100" and "Mountain-500" perform poorly online (Table 5), indicating a stronger preference for premium products among online

shoppers.

Table 5

Online least popular product

Sum of SalesQuantity		Column Labels	
Row Labels		Online	Grand Total
Mountain-100 Silver, 48		36	36
Mountain-500 Silver, 44		39	39
Mountain-500 Black, 52		41	41
Mountain-100 Silver, 42		42	42
Mountain-500 Silver, 40		45	45

The relationship between popularity and revenue shows that high sales quantities do not always result in high revenue, as it depends on cost structure and pricing. High-end products like "Mountain-200" and "Road-650 Red, 48" generate substantial revenue with moderate sales due to higher prices, while popular low-cost items like "AWC Logo Cap" contribute less. Similarly, cost dynamics reveal that popular product like "Hitch Rack - Bike" have lower costs, while higher-cost items like "Mountain-200" still maintain strong sales (See Table 6 and Figure 30), indicating both low-cost and premium products can achieve popularity.

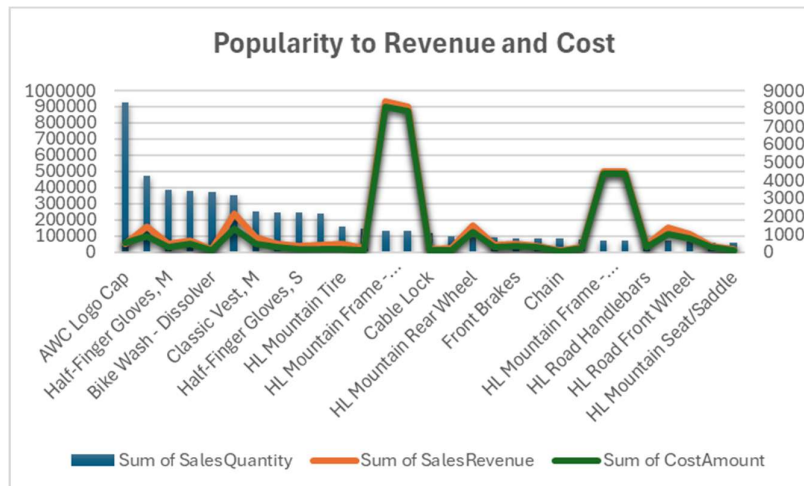
Table 6:

Most popular products to revenue and cost

Row Labels	Sum of SalesQuantity	Sum of SalesRevenue	Sum of CostAmount
AWC Logo Cap	8311	\$ 51,229.45	\$ 57,531.24
Water Bottle - 30 oz.	6815	\$ 28,654.16	\$ 12,718.83
Sport-100 Helmet, Blue	6743	\$ 165,406.62	\$ 88,240.92
Long-Sleeve Logo Jersey, L	6592	\$ 198,754.98	\$ 253,741.24
Sport-100 Helmet, Black	6532	\$ 160,869.52	\$ 85,479.71
Sport-100 Helmet, Red	6266	\$ 157,772.39	\$ 81,998.76
Classic Vest, S	4247	\$ 156,398.07	\$ 100,862.00
Patch Kit/8 Patches	3865	\$ 8,232.60	\$ 3,310.37
Short-Sleeve Classic Jersey, XL	3864	\$ 129,145.57	\$ 160,635.37
Long-Sleeve Logo Jersey, M	3636	\$ 115,249.21	\$ 139,958.00
Half-Finger Gloves, M	3464	\$ 54,545.49	\$ 31,727.82
Full-Finger Gloves, L	3378	\$ 69,943.21	\$ 52,936.30
Bike Wash - Dissolver	3319	\$ 18,406.97	\$ 9,868.38
Women's Mountain Shorts, S	3296	\$ 137,164.13	\$ 86,277.08
Women's Mountain Shorts, L	3244	\$ 136,774.02	\$ 84,915.92
Hitch Rack - 4-Bike	3166	\$ 237,096.16	\$ 142,090.08
Mountain Tire Tube	3095	\$ 15,444.05	\$ 5,776.20
Long-Sleeve Logo Jersey, XL	2980	\$ 95,611.20	\$ 114,707.05
Mountain-200 Black, 38	2977	\$ 4,400,592.80	\$ 3,727,148.33

Figure 30

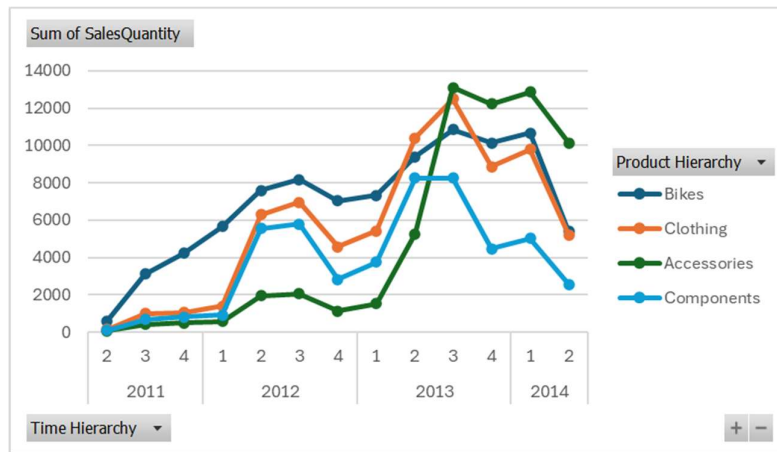
Popularity to revenue and cost



Over time, most categories peaked in popularity in 2013 and then declined in 2014. Sales show seasonality, peaking in Q3, likely due to cycling season or holidays (See Figure 31).

Figure 31

Overall popularity trend



"Accessories" experienced rapid online sales growth starting in 2013, surpassing in-store sales within a year, indicating a shift to online shopping for smaller items (See Figure 32 and 33).

Figure 32

Online popularity trend

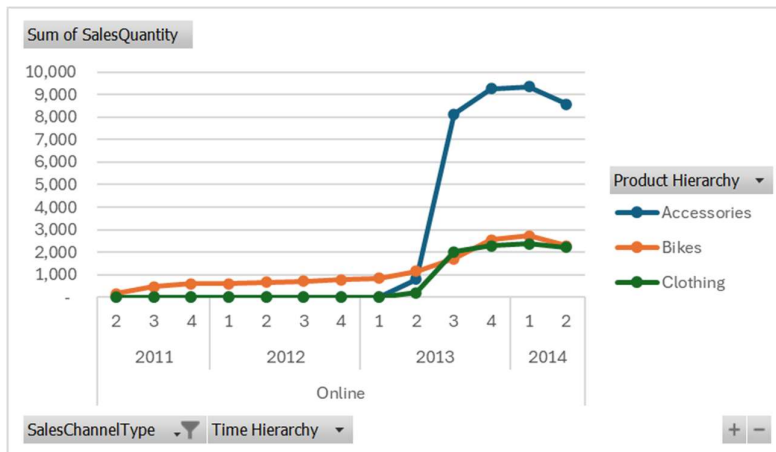
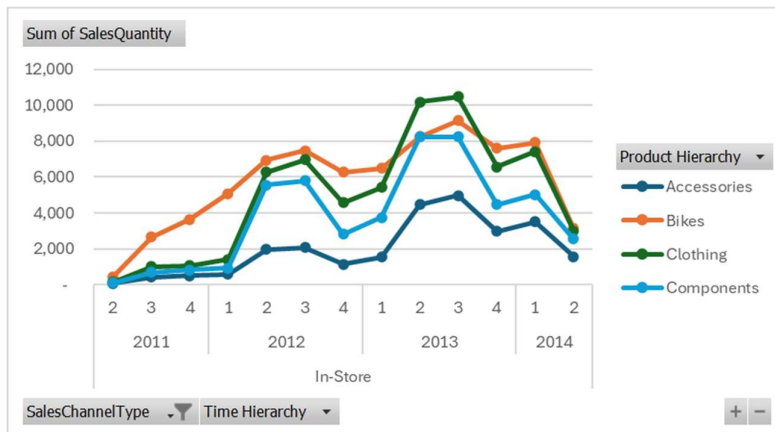


Figure 33

In-store popularity trend



Regionally, the US dominates sales across all categories, followed by Canada, highlighting strong North American demand (See Figure 34).

Figure 34

Popularity by region



6.3 Question 2: Which products are most/ least profitable either instore or online and how does profitability impact revenue/cost of the products?

6.3.1 Cube Usage

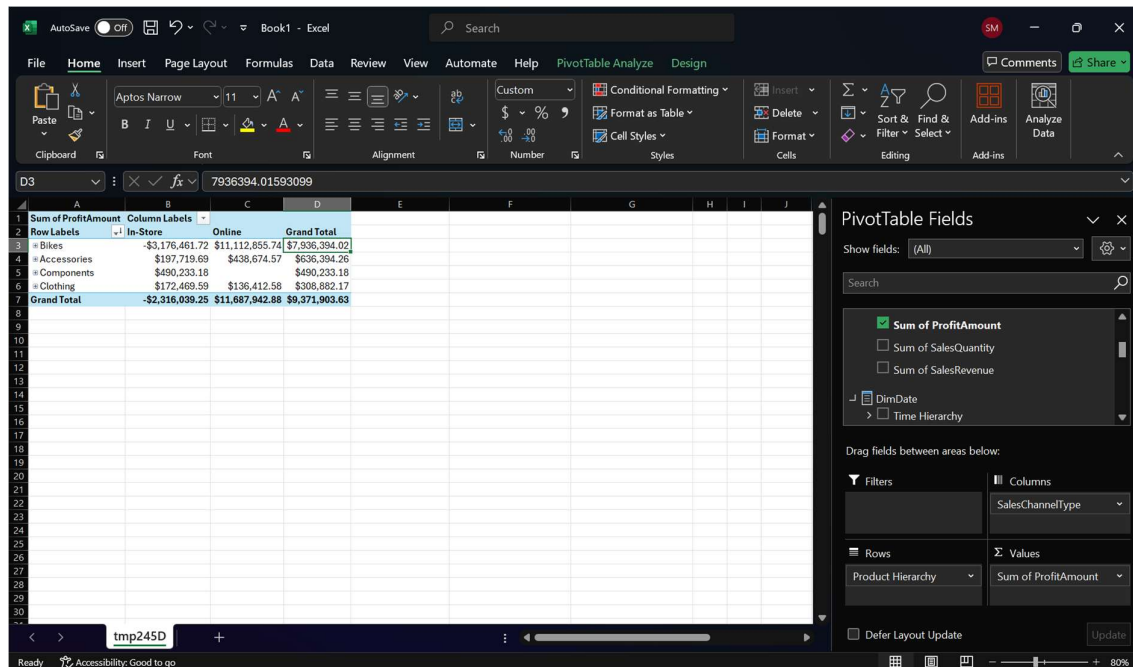
The question was also broken down to be analysed in 3 parts. For answering this question, “Profitability” has been defined by metric of Profit, which is defined by deducting Cost of Goods Sold from Sales Revenue (*Sum of ProfitAmount*).

1. Product Profitability Analysis:

- **Setup:** The analysis started by setting up the PivotTable with SalesChannelType (In-Store and Online) as columns and the Product Hierarchy as rows, with the values showing the Sum of ProfitAmount.
- **Drill-Down:** The Product Hierarchy was drilled down to the Product Name level for detailed insights.
- **Sorting:** Products were sorted by Profitability to identify the most profitable (largest to smallest) and least profitable (smallest to largest) items across different sales channels (See Figure 35).

Figure 35

Pivot Table selection and results for Profitability Analysis (Product Categories)



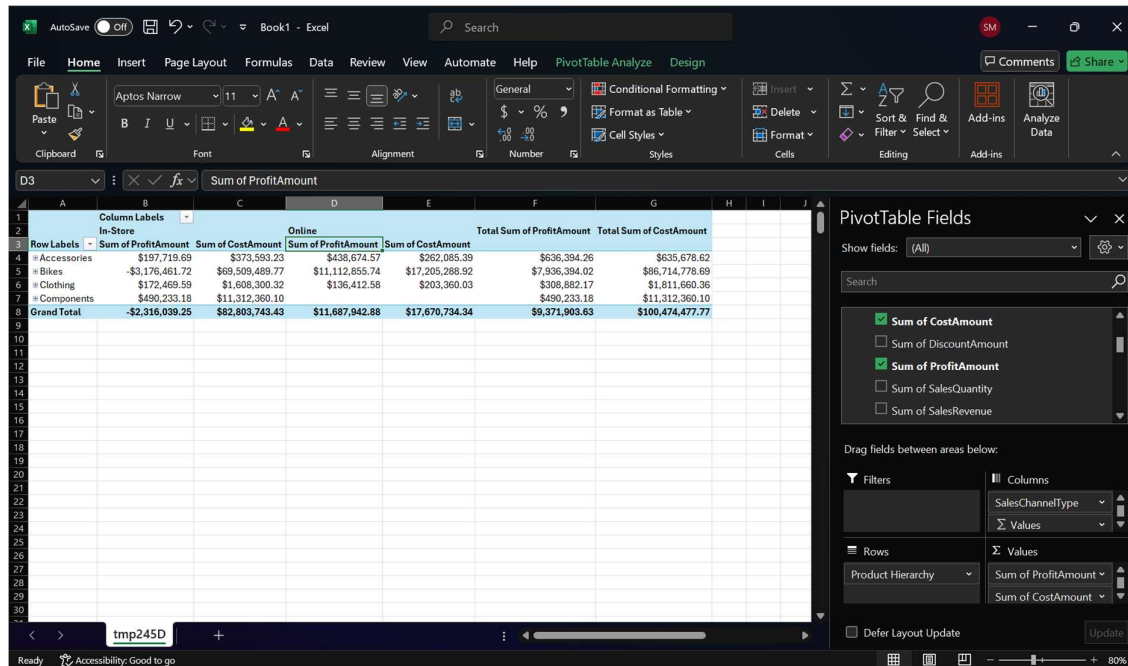
2. Impact on Sales Revenue and Cost:

- **Setup:** A new PivotTable was set up with columns for SalesChannelType, rows for Product Name, and values for Sum of ProfitAmount, Sum of SalesRevenue and Sum of CostAmount (See Figure 36) is used to analyse the impact on revenue and cost.
- **Charts:**

- **Revenue vs. Profitability:** A combo chart was created with bars for Profitability and a line for Sales Revenue, sorted by Profitability to assess the correlation between product popularity and revenue.

Figure 36

Pivot Table selection and results for Impact Analysis on Profitability (Cost Amount)



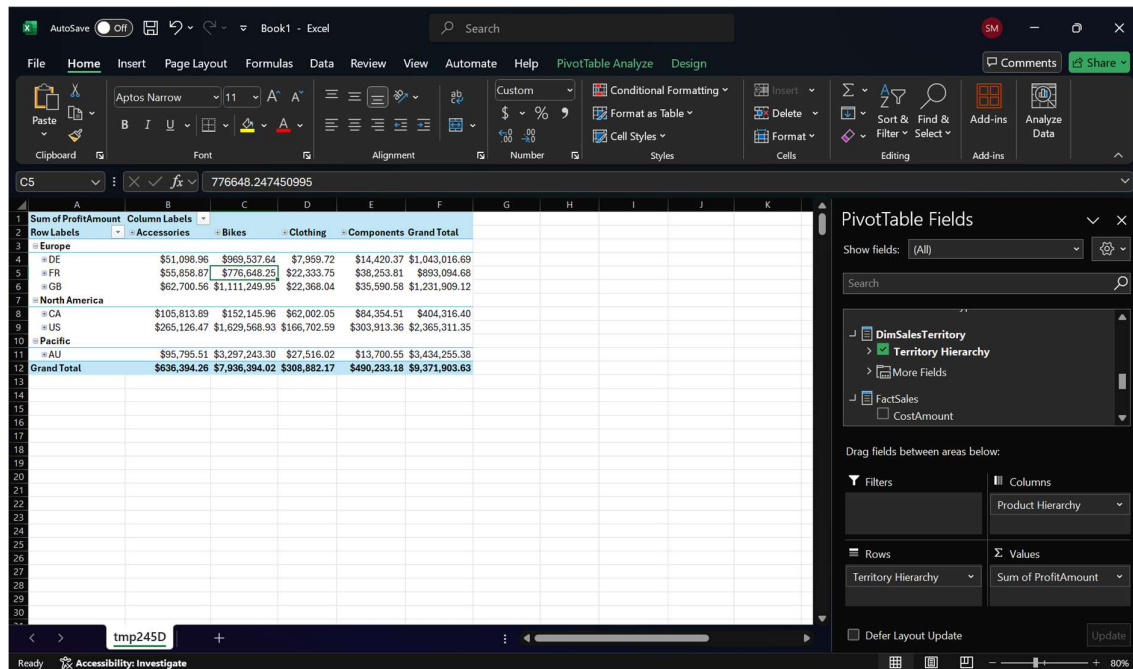
- **Cost vs. Profitability:** Another combo chart displayed bars for Profitability and a line for Cost, also sorted by Profitability, to evaluate how costs relate to product sales.

3. Trends Over Time and Location:

- **Time Analysis:** For time trends, the columns were set to the Product Hierarchy and rows to the Time Hierarchy, with values for Sum of ProfitAmount. A Line with Markers chart illustrated trends by Quarter.

Figure 37

Pivot Table selection and results for Location Analysis on Profitability (Product Categories)



- **Location Analysis:** For geographical insights, the columns included the Product Hierarchy, rows were set to the Territory Hierarchy (drilled down to Country), and values were Sum of ProfitAmount. A Clustered Column chart showed sales performance across different countries (See Figure 37).

6.3.2 Findings

The "Bikes" category contributes about 85% of total profitability, primarily driven by strong online sales (See Table 7).

Table 7

Profitability by category

Sum of ProfitAmount	Column Labels		
Row Labels	In-Store	Online	Grand Total
Bikes	-\$3,176,461.72	\$11,112,855.74	\$7,936,394.02
Accessories	\$197,719.69	\$438,674.57	\$636,394.26
Components	\$490,233.18		\$490,233.18
Clothing	\$172,469.59	\$136,412.58	\$308,882.17
Grand Total	-\$2,316,039.25	\$11,687,942.88	\$9,371,903.63

Within this category, "Mountain Bikes" and "Road Bikes" are the top performers (Table 8).

Table 8*Profitability by "Bikes" subcategory*

Sum of ProfitAmount	Column Labels		
Row Labels	In-Store	Online	Grand Total
Bikes			
Mountain Bikes	\$552,381.52	\$4,355,660.37	\$4,908,041.89
Road Bikes	-\$2,491,248.25	\$5,302,322.67	\$2,811,074.42
Touring Bikes	-\$1,237,594.99	\$1,454,872.70	\$217,277.70

The "Mountain-200" series is particularly profitable across both channels, with models like "Mountain-200 Black, 38" excelling in-store (Table 9) and "Mountain-200 Black, 46" leading online. "Road Bikes," especially "Road-150 Red, 48," also drive online profitability, showcasing the advantage of premium models (Table 10).

Table 9*In-store most profitable products*

Sum of ProfitAmount	Column Labels	
Row Labels	In-Store	Grand Total
Mountain-200 Black, 38	\$107,231.45	\$107,231.45
Mountain-200 Black, 42	\$79,791.00	\$79,791.00
Mountain-200 Silver, 38	\$78,631.37	\$78,631.37
Hitch Rack - 4-Bike	\$70,366.72	\$70,366.72
Mountain-200 Black, 46	\$69,499.55	\$69,499.55

Table 10*Online most profitable products*

Sum of ProfitAmount	Column Labels	
Row Labels	Online	Grand Total
Mountain-200 Black, 46	\$597,241.14	\$597,241.14
Mountain-200 Black, 42	\$594,425.58	\$594,425.58
Mountain-200 Silver, 38	\$585,153.57	\$585,153.57
Mountain-200 Silver, 46	\$567,040.79	\$567,040.79
Mountain-200 Black, 38	\$566,213.02	\$566,213.02
Mountain-200 Silver, 42	\$548,687.65	\$548,687.65
Road-150 Red, 48	\$474,150.84	\$474,150.84
Road-150 Red, 62	\$472,743.87	\$472,743.87
Road-150 Red, 52	\$424,906.69	\$424,906.69
Road-150 Red, 56	\$415,057.86	\$415,057.86

Conversely, "Touring Bikes" are the least profitable, with "Touring-3000 Blue, 58" underperforming in-store (Table 11), and lower-profitability items from "Clothing" and

"Accessories" like "Racing Socks, L," and "Patch Kit/8 Patches" are underperforming online (Table 12).

Table 11

In-store least profitable products

Sum of ProfitAmount	Column Labels	
Row Labels	In-Store	Grand Total
Touring-1000 Yellow, 60	-\$259,044.21	-\$259,044.21
Road-250 Black, 44	-\$243,353.52	-\$243,353.52
Touring-1000 Yellow, 46	-\$218,141.44	-\$218,141.44
Road-250 Black, 48	-\$209,487.79	-\$209,487.79
Road-650 Red, 44	-\$173,182.90	-\$173,182.90

Table 12

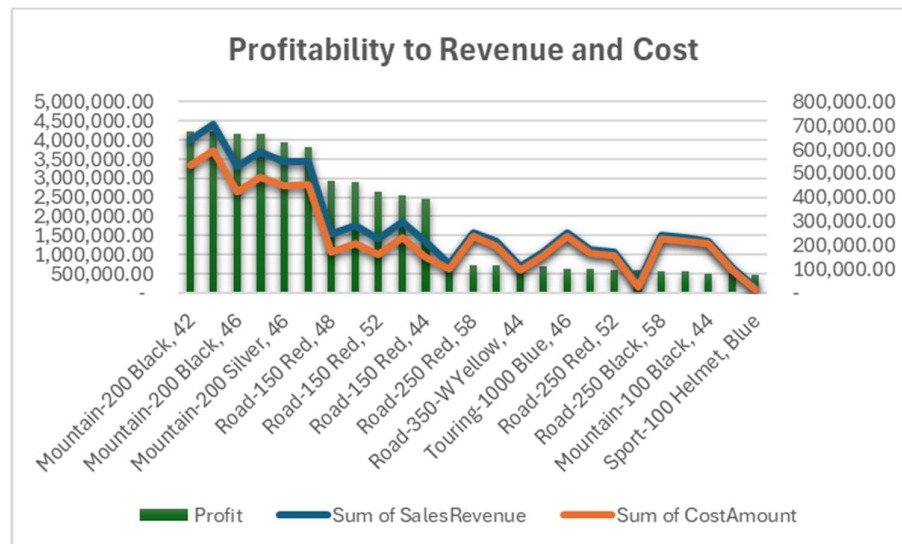
Online least profitable products

Sum of ProfitAmount	Column Labels	
Row Labels	Online	Grand Total
Racing Socks, L	\$1,519.48	\$1,519.48
Racing Socks, M	\$1,677.05	\$1,677.05
Bike Wash - Dissolver	\$4,518.84	\$4,518.84
AWC Logo Cap	\$4,528.26	\$4,528.26
Patch Kit/8 Patches	\$4,574.30	\$4,574.30

Table 13 and figure 38 reveal that while higher sales revenue often correlates with higher profits, the relationship is not always linear. Products like "Road-150 Red, 48" demonstrate strong revenue and profitability due to effective cost management. However, other products with similar revenue, such as "Road-250 Red, 58," have lower profits due to higher costs, indicating that managing production and operational costs is crucial for maximizing profits, particularly in maintaining profitability across different price segments.

Table 13*Most profitable products to revenue and cost*

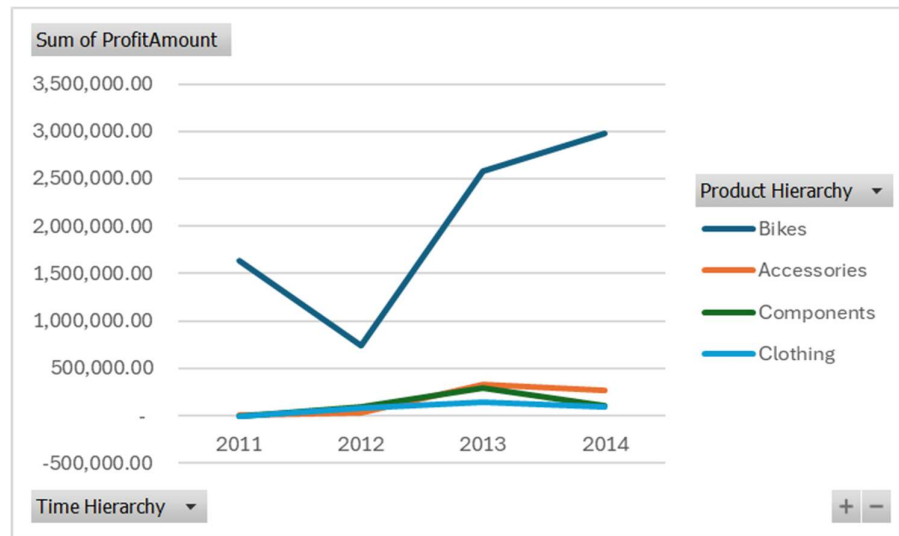
Row Labels	Sum of SalesRevenue	Sum of CostAmount	Profit
Mountain-200 Black, 42	\$ 4,009,494.76	\$ 3,335,278.18	\$674,216.58
Mountain-200 Black, 38	\$ 4,400,592.80	\$ 3,727,148.33	\$673,444.47
Mountain-200 Black, 46	\$ 3,309,673.22	\$ 2,642,932.52	\$666,740.69
Mountain-200 Silver, 38	\$ 3,693,678.03	\$ 3,029,893.08	\$663,784.94
Mountain-200 Silver, 46	\$ 3,434,256.94	\$ 2,804,612.81	\$629,644.13
Mountain-200 Silver, 42	\$ 3,438,478.86	\$ 2,827,393.96	\$611,084.90
Road-150 Red, 48	\$ 1,540,803.06	\$ 1,070,448.04	\$470,355.02
Road-150 Red, 62	\$ 1,769,096.69	\$ 1,302,776.52	\$466,320.17
Road-150 Red, 52	\$ 1,415,563.61	\$ 994,452.74	\$421,110.87
Road-150 Red, 56	\$ 1,847,818.63	\$ 1,441,739.35	\$406,079.28
Road-150 Red, 44	\$ 1,340,419.94	\$ 948,855.57	\$391,564.38
Touring-1000 Blue, 54	\$ 743,353.03	\$ 612,040.35	\$131,312.67
Road-250 Red, 58	\$ 1,587,008.18	\$ 1,470,980.71	\$116,027.47
Road-250 Red, 48	\$ 1,348,759.50	\$ 1,233,254.56	\$115,504.94

Figure 38*Profitability to revenue and cost*

Profitability trends vary over time and by region. The "Bikes" category saw a steady upward trend in profitability from 2012 to 2014 (Figure 39), especially in online sales, highlighting a growing preference for online purchases.

Figure 39

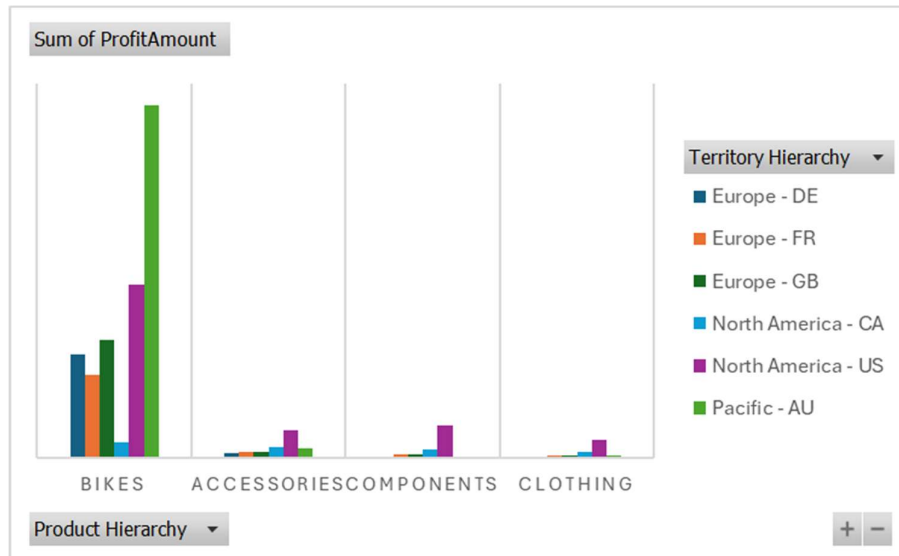
Profit trend by category overtime



Regionally, Australia (AU) leads in overall profitability (Figure 40), driven by strong online sales.

Figure 40

Profitability by region



In contrast, North America (US) has the highest online sales but faces losses in in-store sales (Table 14), highlighting the need for different strategies for different markets and sales channels.

Table 14*Profitability by region and sales channel*

Sum of ProfitAmount	Column Labels									
Row Labels	Europe		North America		Pacific				Grand Total	
	In-Store	Online	In-Store	Online	In-Store	Online				
Accessories	\$ 43,038.12	\$ 126,620.28	\$ 145,706.17	\$ 225,234.19	\$ 8,975.40	\$ 86,820.10			\$ 636,394.26	
Bikes	-\$ 519,722.25	\$ 3,377,158.08	-\$ 2,524,594.30	\$ 4,306,309.18	-\$ 132,145.18	\$ 3,429,388.48			\$ 7,936,394.02	
Clothing	\$ 23,739.06	\$ 28,922.45	\$ 147,982.19	\$ 80,722.44	\$ 748.33	\$ 26,767.68			\$ 308,882.17	
Components	\$ 88,264.76		\$ 388,267.87		\$ 13,700.55				\$ 490,233.18	
Grand Total	-\$ 364,680.31	\$ 3,532,700.81	-\$ 1,842,638.06	\$ 4,612,265.81	-\$ 108,720.88	\$ 3,542,976.26			\$ 9,371,903.63	

6.4 Question 3: Which products would your team recommend to AW to target with sales and marketing promotions and why?

6.4.1 Cube Usage

To answer this question, we referred to the insights gained from the analyses of Question 1 and Question 2.

To better analyse it, we have also added two new measures, **ProfitMargin** and **DiscountImpact**.

To determine which products should be targeted for sales and marketing promotions, the following detailed approach was used, combining PivotTable analysis in Excel and SQL queries:

1. Identifying High-Profit Categories

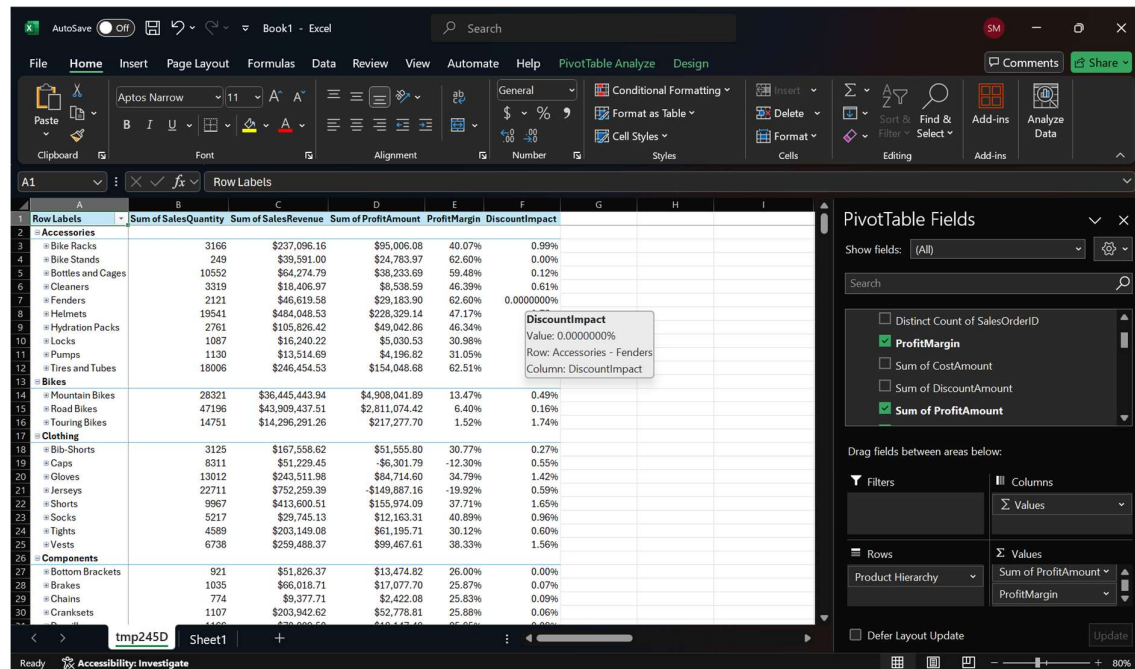
- **Setup:** A Pivot Table was created to analyse profit margins across product categories. The rows were set to the Product Hierarchy, and values included Profit Margin, Discount Impact, Sales Revenue, and Sales Quantity. This setup gave an initial overview of profit margins across different product categories and sub-categories.
- **Drill-Down:** The analysis was drilled down from product categories to specific sub-categories to identify high-margin products in detail. This helped pinpoint the most profitable categories and products (See Figure 41).

2. Analysing High-Performing Products Against Discount Impact:

- **Setup:** Another Pivot Table was configured to examine the relationship between high sales performance and discount impact. rows were set to Product Hierarchy, and values included Sales Quantity, Sales Revenue, Profit Margin, and Discount Impact (See Figure 41).

Figure 41

Pivot table used for overall analysis for parts of answer 3



- Drill-Down: Products with strong sales performance and minimal discount impact were further analysed by drilling down into sub-categories. This detailed view provided insights into which products excel under current discount strategies and which could benefit from additional promotional efforts.

3. Bundling Strategy for Profit Optimization:

- Setup: A Pivot Table was also used to identify profit margins by drilling down from products. This helped in understanding the profitability of individual products and categories.
- SQL Query Analysis: An additional question was addressed through SQL queries in SSMS to gain insights into purchasing behavior. The queries identified frequently bought-together products and product categories commonly paired in bundles. This analysis aimed to provide Adventure Works (AW) with data-driven recommendations for promoting specific product combinations and enhancing cross-selling strategies. The goal was to increase customer satisfaction and drive revenue growth through targeted promotions. This question, the process, and its findings have been further described in the Appendix A.
- Bundling Strategy: A bundling strategy was developed based on the insights from the SQL queries. High-margin products were combined with lower-margin items to create

promotional bundles. This approach sought to balance the profit margins of the bundled products and optimize overall profitability. Detailed findings and additional insights are provided in the Appendix.

This comprehensive approach supports effective decision-making for promotional strategies by focusing on high-profit categories, evaluating the impact of discounts on sales performance, and implementing strategic bundling based on purchasing behaviour analysis.

6.4.2 Findings

1. **Focus on High-Profit Margin Accessories:** High-margin accessories like “Tires and Tubes” with 63% profit margin (See Table 15) should be the focus of aggressive promotions, particularly online, where demand for such essential items has seen consistent growth. Targeting regions like the United States (US) through digital marketing campaigns can tap into the rising interest in bike maintenance products. Tires and Tubes, being necessities for cyclists, have wide market appeal. Additionally, with a discount impact close to 0%, offering modest discounts won’t affect profitability significantly, allowing for flexible promotions that enhance sales without sacrificing margins.

Table 15

Highest profit margin by subcategory

Row Labels	Sum of SalesQuantity	ProfitMargin	Sum of ProfitAmount	DiscountImpact
Accessories				
⊕ Fenders	2121	63%	\$29,183.90	0.00%
⊕ Bike Stands	249	63%	\$24,783.97	0.00%
⊕ Tires and Tubes	18006	63%	\$154,048.68	0.00%
⊕ Bottles and Cages	10552	59%	\$38,233.69	0.12%
⊕ Helmets	19541	47%	\$228,329.14	0.75%
⊕ Cleaners	3319	46%	\$8,538.59	0.61%
⊕ Hydration Packs	2761	46%	\$49,042.86	0.48%
⊕ Bike Racks	3166	40%	\$95,006.08	0.99%
⊕ Pumps	1130	31%	\$4,196.82	0.10%
⊕ Locks	1087	31%	\$5,030.53	0.15%

2. **Promote High-Performing Bikes:** “Road Bike” and "Mountain Bikes" are the highest contributors to company revenue, with a total of \$80.4 million (Table 16), should be a focal point in regions like North America (US). Timing promotions for peak sales periods, such as Q3, can maximize their impact. AdventureWorks can further drive sales

by offering region-specific promotions like financing, loyalty rewards, or seasonal discounts. The discount impact of Mountain Bikes remains low (0.49%), allowing for strategic promotions without eroding profits.

Table 16

Highest revenue by subcategory

Row Labels	Sum of SalesQuantity	Sum of SalesRevenue	ProfitMargin	DiscountImpact
Bikes				
⊕ Road Bikes	47196	\$43,909,437.51	6%	0.16%
⊕ Mountain Bikes	28321	\$36,445,443.94	13%	0.49%
⊕ Touring Bikes	14751	\$14,296,291.26	2%	1.74%
⊕ Components	49044	\$11,802,593.29	4%	0.04%
⊕ Clothing	73670	\$2,120,542.52	15%	0.99%
⊕ Accessories	61932	\$1,272,072.88	50%	0.53%
Grand Total	274914	\$109,846,381.40	9%	0.48%

3. **Bundle Negative Margin Products with High-Margin Items:** To offset losses, products with negative profit margins, like Jerseys (-20%), can be bundled with high-margin products like the Sport-100 Helmet (47%) (See Table 17). These bundles target cycling enthusiasts and can improve profitability by increasing the average order value and clearing slow-moving inventory.

Table 17

Profit Margin of “Helmets” and “Jerseys” subcategory

Row Labels	Sum of SalesQuantity	ProfitMargin
Accessories		
⊕ Helmets	19541	47%
Clothing		
⊕ Jerseys	22711	-20%
Grand Total	42252	6%

AdventureWorks should prioritize online sales, where bikes and accessories show strong revenue growth. By expanding digital marketing efforts and optimizing the online shopping experience, they can capture this demand and accelerate growth through targeted strategies.

7. Conclusion

The comprehensive analysis of the data warehouse (DW) and the exploration of the SSAS cube has generated key insights that directly support Adventure Works (AW) in promoting the best products to sustain and accelerate business growth. Through detailed examination of sales data, profitability, and product performance across both in-store and online channels, several critical findings emerged that can guide AW's marketing and sales strategies.

Firstly, product popularity was a significant factor in determining which items should be prioritized for promotion. The analysis revealed that bikes, particularly high-end models such as the "Mountain-200" series and accessories like the "Sport-100 Helmet" and "AWC Logo Cap," are consistently popular across both sales channels. The demand for these products is driven by their association with high-performance cycling, and their frequent bundling with complementary items like jerseys and gloves highlights a strong opportunity for cross-selling. Additionally, products such as clothing and accessories dominate the online sales space, indicating a potential for AW to further target these categories with digital marketing campaigns, especially given the growing trend of online shopping.

The profitability analysis highlighted that high-margin products like tires and tubes, with profit margins as high as 63%, offer substantial opportunities for promotional campaigns. These products not only deliver strong profitability but also align with customer demand, making them ideal candidates for bundled promotions alongside lower-margin items like jerseys. By bundling high-margin items with slower-moving products, AW can boost overall profitability while clearing inventory. The analysis also showed that the Mountain-200 bike series is highly profitable across both channels, reinforcing the need to focus promotional efforts on this line to maintain strong sales performance.

Additionally, the data revealed geographical and seasonal trends that can help AW fine-tune its marketing strategies. Sales peak during Q3, coinciding with the cycling season, which presents an opportunity for seasonal marketing campaigns that target cycling enthusiasts. Moreover, the United States emerged as the dominant region for sales, underscoring the importance of region-specific promotions that capitalize on local demand patterns.

The cube analysis also identified the relationship between popularity and revenue. It was observed that while popular items drive sales, their profitability depends on the cost structure and pricing strategies. Products like the "Mountain-200" generate high revenue with moderate sales due to premium pricing, whereas popular low-cost items like caps contribute less to overall revenue. This insight suggests that AW should balance its promotional efforts between high-revenue premium products and high-volume lower-cost items to maximize overall profitability.

The findings from the SSAS cube provided Adventure Works with actionable insights into customer purchasing behaviours, including frequently purchased product bundles and category pairs. For instance, bikes and components are often bought together, and high-margin products like helmets are frequently bundled with lower-margin items. These insights point to the value of implementing **strategic bundling** and cross-promotion efforts to encourage customers to purchase related items, thereby increasing the average order value and improving the overall customer experience.

In conclusion, the DW and cube analysis provide a data-driven roadmap for Adventure Works to promote the most profitable and popular products effectively. By leveraging these insights, AW can refine its marketing campaigns, optimize product bundles, and tailor promotions to align with customer preferences, seasonal trends, and regional demand. These strategies will not only sustain AW's current growth but also create new opportunities for expansion and increased profitability, helping the company remain competitive in a dynamic market.

Appendix A

Visual Analysis of Product Bundles and Category Pairs

This appendix presents visualizations based on the analysis of customer purchasing patterns. The figures highlight both individual products and product categories that are frequently purchased together, offering insights into Adventure Works' bundling opportunities for marketing campaigns and cross-selling strategies.

A. Introduction

The goal of this analysis is to gain insights into the purchasing behavior of customers by identifying both individual products that are frequently bought together and product categories that tend to be paired in bundles. The purpose is to provide Adventure Works (AW) with data-driven recommendations for promoting specific product combinations and improving cross-selling strategies, which can ultimately increase customer satisfaction and drive revenue growth.

The analysis is divided into two parts:

1. Investigating Products with the Most Times Purchased Together: This part focuses on determining which individual products are involved in high-frequency bundles. By identifying the most frequently paired products, AW can target these specific items in marketing campaigns, bundling promotions, or discounts. These products could be promoted either as individual upsell opportunities or as part of larger bundles to encourage customers to buy related products together.
2. Analysing Which Categories Go Well Together: In addition to focusing on specific products, the second part of the analysis explores product categories that are often purchased together. This investigation helps understand broader customer buying patterns across product lines. By examining which categories are frequently bundled, such as bikes and components or accessories and clothing, AW can optimize its promotional strategies across entire categories rather than individual products. This helps ensure that products from complementary categories are marketed as a cohesive unit, encouraging customers to purchase related items in a single transaction.

By understanding both product-level and category-level bundling patterns, AW can make informed decisions about which products and categories to target for cross-promotion,

marketing campaigns, and sales initiatives. These insights are crucial for sustaining growth and enhancing the overall shopping experience for AW's customers, whether online or in-store.

B. Process

The analysis involved both SQL queries for extracting data and Excel for further processing and visualization.

Step 1: Extract Data Using SQL Queries

1. Bundle Query for Product Category: This query paired products based on their categories and counted how often they were purchased together (See Figure 1, Appendix A).

Figure 1

Bundle Query for Product Category

```
SELECT
    SF1.ProductID AS ProductID1,
    P1.ProductCategory AS ProductCategory1,
    SF2.ProductID AS ProductID2,
    P2.ProductCategory AS ProductCategory2,
    COUNT(*) AS TimesPurchasedTogether
FROM
    dbo.FactSales SF1
JOIN
    dbo.FactSales SF2 ON SF1.SalesOrderID = SF2.SalesOrderID
    AND SF1.ProductID < SF2.ProductID -- Avoid self join on the same product
JOIN
    dbo.DimProduct P1 ON SF1.ProductID = P1.ProductID
JOIN
    dbo.DimProduct P2 ON SF2.ProductID = P2.ProductID
GROUP BY
    SF1.ProductID, P1.ProductCategory, SF2.ProductID, P2.ProductCategory
ORDER BY
    TimesPurchasedTogether DESC;
```

2. Bundle Query for Product Name: This query worked similarly but focused on individual product names (See Figure 2, Appendix A).

Figure 2

Bundle Query for Product Name

```
SELECT
    SF1.ProductID AS ProductID1,
    P1.ProductName AS ProductName1,
    SF2.ProductID AS ProductID2,
    P2.ProductName AS ProductName2,
    COUNT(*) AS TimesPurchasedTogether
FROM
    dbo.FactSales SF1
JOIN
    dbo.FactSales SF2 ON SF1.SalesOrderID = SF2.SalesOrderID
    AND SF1.ProductID < SF2.ProductID -- Avoid self join on the same product
JOIN
    dbo.DimProduct P1 ON SF1.ProductID = P1.ProductID
JOIN
    dbo.DimProduct P2 ON SF2.ProductID = P2.ProductID
GROUP BY
    SF1.ProductID, P1.ProductName, SF2.ProductID, P2.ProductName
ORDER BY
    TimesPurchasedTogether DESC;
```

Step 2: Process Data in Excel

Product Analysis – Products with Most Times Purchased Together

1. Step 1: Combine Product 1 and Product 2 with Times Purchased Together
 - Copy ProductName1 into one column.
 - In the next column, copy Times Purchased Together.
 - Below that, copy ProductName2 into the first column and Times Purchased Together into the second column.
 - This step created a unified list of all products and how many times they appeared in bundles.
2. Step 2: Calculate Total "Times Purchased Together" for Each Product
 - Select the combined list (with product names and their corresponding "Times Purchased Together").
 - Insert a Pivot Table.
 - Drag ProductName to the Rows section.
 - Drag Times Purchased Together to the Values section, ensuring the value is set to Sum.

3. Step 3: Sort the Results

- Sort the values in descending order by Times Purchased Together to find the products most frequently purchased together.

Category Pairs – Analysing Which Categories Go Well Together

1. Step 1: Combine Product Categories

- In the "Bundle Query for Product Category" sheet, create a new column.
- In this column, use the formula to combine the two categories into a single pair

=IF(C2<E2, C2 & " + " & E2, E2 & " + " & C2)

- Copy the formula down for the entire dataset to standardize all product category pairs.

Step 2: Create a Pivot Table to Summarize the Category Pairs

- Select the combined data, including the new category pairs column and the "Times Purchased Together" column.
- Insert a Pivot Table.
- Drag the combined product category pairs to the Rows section and Times Purchased Together to the Values section, ensuring the value is set to Sum.

Step 3: Sort the Pivot Table

- Sort the results by Times Purchased Together in descending order to see which category pairs were purchased together the most.

C. Findings:

The analysis of product bundles based on the product names reveals that certain products are frequently purchased together, providing insights into customer preferences. The most commonly bundled product is the AWC Logo Cap, which was paired with other products 38,553 times. Other high-frequency bundled products include various helmets such as the Sport-100 Helmet, Blue (35,657 times) and the Sport-100 Helmet, Black (35,021 times). These helmets are essential accessories for biking and are often sold in conjunction with related items like jerseys and gloves (see Figure 3, Appendix A).

Figure 3

Most Frequently Purchased Products in Bundles

Product Name	Sum of Times Purchased Together
AWC Logo Cap	38553
Sport-100 Helmet, Blue	35657
Sport-100 Helmet, Black	35021
Long-Sleeve Logo Jersey, L	34774
Sport-100 Helmet, Red	33942
Long-Sleeve Logo Jersey, M	28457
Half-Finger Gloves, M	25048
Water Bottle - 30 oz.	24816
Long-Sleeve Logo Jersey, XL	24227
Half-Finger Gloves, S	18042
Mountain-200 Black, 38	17611
LL Road Frame - Black, 52	17533
Bike Wash - Dissolver	16917
Mountain-200 Black, 42	16728
Short-Sleeve Classic Jersey, XL	16593
Mountain-200 Silver, 38	16588
Mountain-200 Silver, 42	16362
ML Mountain Handlebars	16206
Classic Vest, S	16151
Mountain-200 Silver, 46	16123

The Long-Sleeve Logo Jersey, L (34,774 times) and Long-Sleeve Logo Jersey, M (28,457 times) also appear frequently in bundles, indicating that clothing, especially jerseys, plays a significant role in customers' purchasing decisions, likely as part of cycling outfits. Accessories such as the Water Bottle - 30 oz. (24,816 times) and Half-Finger Gloves, M (25,048 times) are other notable items that are bundled regularly, often as add-ons to larger purchases like bikes or helmets (refer to Figure 3 for additional details on these products).

The findings from the category pairs further reinforce these product trends, with the most common category combinations being Bikes + Components (212,162 times) and Bikes + Bikes (149,196 times). This shows that customers frequently purchase bikes in combination with either their components or other bikes, highlighting the importance of offering deals or discounts when multiple bikes are purchased (see Figure 4, Appendix A for the breakdown of category pairs).

Figure 4

Product Category Pairs with the Highest Times Purchased Together

Category pairs	Sum of Times Purchased Together
Bikes + Components	212,162
Bikes + Bikes	149,196
Bikes + Clothing	129,139
Clothing + Components	117,595
Components + Components	114,809
Accessories + Bikes	70,920
Accessories + Clothing	59,911
Clothing + Clothing	52,067
Accessories + Components	50,419
Accessories + Accessories	35,920
Grand Total	992,138.00

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